THE OMNI-COMPASS LIVE ENGINE MANUAL

How to Read the Live Engine:
Its Monitors, Gauges and Knobs,
Native beside Omni-Compass

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THE OMNI-COMPASS LLC
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Disclaimer. The software is provided “as is”, without warranty of any kind. This book is the reading manual of the live engine page and carries no benchmark results of its own; every published result carries its evidence class. A modelled result is a statement about a model; a software benchmark is a statement about the software it ran on; only a hardware meter speaks for hardware. Nothing in this book is a promise of a particular saving on a particular system: the only number that applies to a system is the one its own paired runs produce, on its own receipt.

Set in Liberation Serif and DejaVu Sans Mono, the text at 11 pt on a 16.5 pt line. Built from the repository by `docs/book/build_book.py`; the book and the code change in the same commit.

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Foreword

A governor that cannot be watched cannot be trusted. Every controller a stack already runs has a dashboard somewhere: the autoscaler's replica count, the card's clock and power, the building's setpoints. What none of them had was one wall where the whole body is seen at once, native beside the same body governed, decision by decision, with the record each number came from one click away. The live engine is that wall, and this book is how to read it.

I asked for it to be a monitor and not a drawing: every record on it made again by the repository's own code, every raw file checked against its own SHA-256 list before a byte of it is drawn, every word on it the run's own reading and never one drawn from the stream. So the page can be trusted the way a meter is trusted: it shows what the record holds, and when the record has nothing yet (a cloud set still running, a chip run not yet started) it says so and shows nothing in its place.

Read the key first. Then stand at the wall. The switch and the engine come first, the four signals of the living organism next, and every reading that went against native is raised before anything that went for it. If a tile reads LOSS, this book tells you where its record is and what to do: the row stays, its cause is found (our wiring first, native's setup second, the scoring third), and the test runs again. Nothing on the page is pushed toward GAIN; the only thing forced is the mechanism.

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Preface

This is the Live Engine Manual, the third book: how to read the live engine page, the monitor of the engine and of the benchmark runs GitHub ran. It is for the operator at the dashboard: which tile means what, which knob does what, where every number comes from, and how the mechanism of action of Omni-Compass against native shows on the monitors. It explains neither how the engine is built (the Owner's Manual) nor how it is wired onto a stack (the Operator's Manual).

The book is in eight parts, and it can be read in two ways.

You are	Read
The operator on shift, at the wall	the Quick Reference card; Part Two (the wall and the switch); the chapter on a tile that reads LOSS in Part Seven; the troubleshooting appendix
CTO, architect, head of platform	the front matter; Part Three (the whole organism and the benefit board); Part Seven (the mechanism side by side; what the page does not tell you)
CEO, CFO, CIO, board member	the Foreword; the Disclosures; the bills and receipts in Part Three; the clouds in Part Four
Site reliability engineer, on call	the Quick Reference card; Part Two; the official run playback in Part Four (the heart and the two tickers); the gauges appendix
Control or plant engineer	Part Six (one cycle, the phase plane, the brain's state, the verdict, the specification); the console in Part Five
Auditor, diligence team	Part One (a monitor, not a drawing); the trace from a tile to its record and the verification chapter; the page's own appraisal
Procurement and counsel	the Disclosures and the papers bound into the back matter

Part One says what the page is and how it is built. **Part Two** is the wall and the switch. **Part Three** is the whole organism and the measured effect. **Part Four** is the evidence stack by stack, from the cloud down to the chip. **Part Five** is the organism live and the console. **Part Six** is the brain. **Part Seven** reads a result off the page. **Part Eight** is the license and how to verify a copy.

Conventions. `Code` is a command, file, anchor or switch exactly as typed. "Native" is the system by itself, amber; "Omni-Compass" is Omni-Compass on top of native, blue; there is no third side. "A section" is a

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numbered part of the page, named by its anchor; "a chapter" is a chapter of this book. A knob marked **LIVE** acts on the live engine; one marked **RECORDED** chooses a record and modifies nothing. Every number carries its evidence class: P a physical meter, L live software, S our own models (modelled); a model is never proof.

The standards every page of this manual keeps. The page, and so this book, keeps the following discipline throughout:

1. **Two sides, always.** Every monitor shows native beside Omni-Compass, the same whole system on the same demand, the same draw of noise, the same injected fault, at the same second. Omni-Compass is never shown standalone and never as a competitor to the controllers it sits on.
2. **Nothing typed in.** Every number is read from a record the repository holds when the page is built or opened; every raw file is matched against its run's own SHA-256 list before it is drawn; the brain, the compass law and the verdict running in the page are the engine's own code, ported and proven equal to the Python and the C++ on every build (`docs/engine/check_live_engine.py`).
3. **The record's own word.** Every word on the page is the reading its record carries under the one key; the page never draws a word from the stream it plays, and never prints "up", "down", "better" or "worse".
4. **Every row, LOSS included.** A reading that went against native is raised first, as an alarm, with its source record and the repository's root-cause document where one exists. Nothing is cut from the page because it went against Omni-Compass.
5. **Nothing read across engine versions.** The page names the engine it stands on and refuses to be built on another; each three-run table on it names its own engine, and a table stands on an older engine only until its runs on the engine that stands land.
6. **Models are not proof.** Every view says its evidence class. The official Kubernetes and database runs are class L, played from their own checked files. The harness, the shipped controller on the toy cluster, the GPU card, the Linux host, the realms table and the grid are class S, our own models, shown beside the real runs and never presented as proof.
7. **Do no harm, in its one meaning.** Every knob on the console acts on both sides alike; what Omni-Compass does with it is held by its guards: no muscle driven below native or past its own limit, nothing taken from one muscle to give to another on the engine that stands, a give-back tried against native first and refused on a LOSS (`DISCLOSURES.md`, the page's section 3).

Prelude: A Night at the Wall

It is a quarter past two. The on-call engineer has the wall open on one screen and the cluster's own console on the other. On the wall the switch is green and reads ON. Under it the four signals of the service: the work waiting, the work done, the decisions past their line, the capacity held; native in amber on the left of each tile, Omni-Compass in blue on the right, both moving on the same demand. The amber line of the work waiting

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climbs as the night's batch starts; the blue line climbs too, then bends sooner, because the governor's ticker (the page's section 4c) shows the pedal pushed at the first decision the compass left its center, and two machines recommended before native's autoscaler had seen its first late sample.

A tile in the alarms row turns red. It reads LOSS: the pod start wait, mean, +13% in one run. The engineer clicks it and lands on the three-run table it came from: run A of the Azure steady set, 33 paired repetitions, the interval −44 to +60 seconds, *no* in the significant column. The red is the run's own word for this run; in the three-run reading it is not yet a reading at all, because the interval includes zero. The row stays on the page, as every row does. The engineer opens the root-cause document the tile links to, `docs/POD_START_WAIT.md`, and reads that a pod's start on a shared host is the test machine's own unpacking of an image, the same on both sides at the same host load. Nothing to do tonight; the record is complete.

At three the batch is done. The blue line of the capacity held eases back a step at a time toward the amber one: the pedal lifted, the lease walking every setting toward native. The engineer scrolls to the page's section 7b, the phase plane, and watches sixty dots drift from the right edge back toward the middle line, native's a few decisions behind. Then back to the wall. Switch green, four signals calm, the alarms row empty but for the one red tile that will stay until its runs B and C land and the three of them are read together. The page's own age, in the last tile, reads twelve minutes.

Introduction: The Page, Its Monitors and What This Manual Covers

The problem the page answers. A supervisory governor sits on top of controllers that each have their own dashboard, and the owner of the stack has to answer one question about it every day: is it doing what the receipts say, right now, on this body, and would I see it if it were not? The receipts are tables; a table is read after the fact. The page is the same record played back in motion, native beside Omni-Compass, with the governor's own decisions and the native controllers' own moves on two tickers side by side, so the mechanism is seen, not inferred.

What the page is made of. Four exporters build its records on the engine that stands: the realms harness (one organism and each muscle alone; `docs/engine/export_harness_trace.py`), the shipped controller on the toy cluster (on top of the autoscaler at the max, and in the cloud's setting; `docs/engine/export_controller_trace.py`), the official runs read back from their raw files and scored again by the repository's own scorer (`docs/engine/export_runs.py`, `tools/live_reps.py`), and the chip and host models with every judged row of the newest receipts (`docs/engine/export_scenarios.py`). A fifth record is the build itself: the engine, the commit, the

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time, and GitHub's own list of the runs in progress and of every run started for the chips and the clouds (`docs/engine/github_ledger.json`).

What runs live in the page. The brain (`omnicompass/core.py`, integrated by the fourth-order Runge–Kutta method), the compass law (`omnicompass/compass_law.py`) and the verdict (`omnicompass/verdict.py`) run in the page itself, ported to the browser and proven equal to the Python and the C++ on every build. They govern a stress-test model of sixty muscles, each on top of its own native controller, and they govern the page's own drawing (the page's section 10). Everything else is a record played back.

What this manual covers. Every section of the page in the page's own order; every gauge, with its side, its aim and its word; every knob of the console, with what it acts on and what it leaves alone; the two tickers that show the mechanism of action; the trace from any tile to the record it came from; and what to do when a tile reads LOSS. It covers the page as built on omni-v13; a later engine renames itself on the page's title block and this book is realigned in the same push.

What this manual does not cover. How the engine is built, its equations beyond the ones the page itself prints, the wiring of Omni-Compass onto any stack, and the benchmark results themselves as a body of evidence: those are the Owner's and the Operator's Manuals and `docs/STATE_OF_PLAY.md`.

The key: GAIN, EVEN and LOSS

Every judged reading in Omni-Compass is one of three words: **GAIN**, **EVEN** or **LOSS**. Better, same and worse are not used anywhere. The word is decided by what the gauge is for, never by the plus or minus sign of its number: who sets the aim differs from one system to the next, and each gauge's aim is written in its preregistration before the runs.

Word	What each letter stands for	Its sense in underwriting	What it means in a table
GAIN	**G** Growth (asset appreciation); **A** Advantage (risk reduction); **I** Improvement (margin expansion); **N** Net-positive (profitable outcome)	surplus and positive variance	the gauge moved toward its declared aim against native: more work done, a shorter wait, fewer misses or failures, or a resource cut
EVEN	**E** Equalized (zero variance); **V** Verified baseline (expected norm); **E** Equilibrium (balanced risk); **N** Neutral (no net change)	the break-even baseline: equilibrium	native and Omni could not be told apart: the same, or no difference beyond the noise; a knob native already runs at the top of its range is EVEN, at its limit on native

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Word	What each letter stands for	Its sense in underwriting	What it means in a table
LOSS	**L** Leakage (capital drain); **O** Off-track (adverse selection); **S** Setback (claim liability); **S** Subpar (negative variance)	actuarial deficit and exposure	the gauge moved away from its declared aim: less work done, a longer wait, more misses or failures

How the sign of a number becomes a word:

- **Time waited, missed or failed** (a response time, time over the line, failed requests): a fall is GAIN, a rise is LOSS.
- **Work done** (work inside the line, throughput, a hit rate): a rise is GAIN, a fall is LOSS.
- **A resource spent or held** (energy, the power drawn and its peaks and swings, machines, memory, processor time, a bill, wear on a part): a fall is a **cut**, and a cut is GAIN. A rise is **power given**: what Omni put in to buy the work and the speed shown beside it. Power given is always shown and never hidden, and it is never called a loss (the founder's order of 12 October 2026: the max rides every muscle's edge; it never shaves). A peak is power held, so a peak that rose is power given, in a data centre and in a building alike (17 October).
- **Work bought with each unit of a resource** (work per energy): a rise is a cut in what each unit of work took, a GAIN; a fall is power given, because it falls when more power is given for the same work.
- So a minus can read GAIN (a response time cut 65%, a bill cut 30%) and a plus can read LOSS (failed requests up).

Some preregistrations name their outcome with a label written before the runs (for example SUPERIOR WITHIN GUARDRAILS in the card protocol). The label stays as it was frozen, and every table shows it with its word first: TOWARD ITS AIMS reads GAIN; BOTH WAYS reads GAIN AND LOSS; AWAY FROM ITS AIMS reads LOSS; SAME AS NATIVE reads EVEN (the same as native); AT ITS LIMIT ON NATIVE reads EVEN (at its limit on native); SUPERIOR WITHIN GUARDRAILS reads GAIN (within the guardrails); ENERGY IMPROVEMENT WITH SERVICE TRADEOFF reads GAIN AND LOSS (energy cut, service traded for it); SERVICE IMPROVEMENT WITH ENERGY TRADEOFF reads GAIN, power given (service GAIN, energy given for it); NONINFERIOR / INCONCLUSIVE reads EVEN (inside the noise, within the guardrails); NOT ESTABLISHED reads EVEN (not established: inside the noise, a guardrail crossed on the mean). A run that breaks its own validity checks reads INVALID and carries no word.

A reading counts only when it holds the same way in all three of our runs, A, B and C, with every 95% interval clear of zero: "GAIN in all three of our runs". The other readings are **EVEN, no difference beyond the noise** (in a run the interval includes zero, so native and Omni could not be told apart: that is the result), **the runs disagree** (clear runs pointing different ways: the test is unstable there and is looked into) and **shown, not judged** (a gauge declared in advance as shown only, such as the knob's own setting).

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Why these three words: they are the words of underwriting and actuarial work, where every outcome is graded as surplus, break-even or deficit, and each letter spells out the same rule. Omni-Compass is there to add (GAIN), never to take (LOSS), and to hand a muscle back to its own native setting where it cannot add (EVEN). Its first rule is do no harm, in the founder's meaning of 19 October 2026: the organism Omni-Compass governs is never left turning out less than its native controllers make it turn out, nor less healthy, and every muscle works only as far as its own limit (`DISCLOSURES.md`, section 3). So **LOSS should never appear**: where it does, the row stays in the table, its cause is found and fixed (our own wiring first, then native's setup, then the scoring) and the test is run again.

Disclosure and disclaimer: GAIN, EVEN and LOSS, and the letters behind them, are The Omni-Compass LLC's own convention for reading its own results. The underwriting senses explain the words; they are not an actuarial, financial or insurance opinion, and no outside body defined, reviewed or approved them. A reading states what our own runs measured against native under the declared aims, not a promise about any other system; every number stands beside its word, so a reader can judge it by any other convention. Run and scored by The Omni-Compass LLC's own code. No outside party has reviewed or approved these results. Preregistered means the rules were written down in our own repository before the runs; nothing was filed with any outside body.

What You Are Looking At

What the live engine is. The live engine is one web page, `docs/engine/OMNI_COMPASS_LIVE_ENGINE.html`, built by the repository's own code (`docs/engine/build_live_engine.py`) from the engine that stands and from the repository's own results. It is a monitor of the engine and of the benchmark runs GitHub ran: every number on it is read from a record the repository holds (a three-run table, a run's raw files, the engine's own audit, the realms table, the models' own readings), and every word on it (GAIN, EVEN, LOSS, power given) is that record's own reading under `tools/readings.py`. The page decides nothing about any result and types nothing in. Where it plays the repository's own models (evidence class S) it says so, and a model is never proof.

What "native" and "Omni-Compass" mean on the page. Two names, two sides, nothing else. *Native* is the whole system running by itself: every autoscaler, scheduler, firmware, thermostat, orchestrator and controller it ships with, every muscle wired and moving by its own machinery. *Omni-Compass* always means Omni-Compass on top of native: the same whole system with all of that still running underneath, and the governor on top of all of it, wired in through the two-way plug. No third side exists anywhere on the page. Native is drawn in amber, Omni-Compass in blue, in every chart, table, tile and number.

What a monitor is. The page is laid out as an operations monitor is: a wall of tiles at the top (the status board), each tile a small copy of a monitor further down the page; then the monitors themselves, section by section, each with its own knobs in a bar at its head ("Controls"). A knob marked *Live* acts on the live engine,

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the engine's own mathematics running in the page; a knob marked *Recorded data* only chooses which record is shown and never modifies it. Every monitor shows native beside Omni-Compass on the same demand, the same seconds, so the one difference left between the two sides is Omni-Compass.

What a gauge is. A gauge is one measured quantity of the system (the work waiting, the work done, a response time, the machines in service, the energy drawn, a bill). On the page each gauge is drawn for both sides with its paired reading: the change from native, its 95% interval, and its word. The word is decided by what the gauge is for (the key, above), never by the sign of its number. A rise in a resource reads *power given*, in green, never a loss.

How to read the page, in order. First the switch and the engine (the master switch at the top of the wall, and the engine's name, digest and commit in the title block). Then the four signals a service is watched by: the work waiting, the work done, the decisions past their line and the capacity held, native beside Omni-Compass. Then every reading that went against native, worst first (the alarms). Then the whole organism (the page's section 2), the measured effect by domain (the page's section 3), the evidence stack by stack from the cloud down to the chip (the page's section 4), the organism live (the page's section 5), the console (the page's section 6) and the brain (the page's section 7). The back matter of the page (disclosures, governing law, contact) says what the page is and is not. This manual follows that order, chapter by chapter.

What this manual is, and is not. It is the interpretation of the page: how to follow the dashboard, read each gauge, turn each knob, and see the mechanism of action of Omni-Compass against native as the monitors show it. It does not explain how the engine is built (the Owner's Manual does), and it does not say how to wire Omni-Compass onto a stack (the Operator's Manual does). Nothing in it decides a result: the records of authority are the repository's result files, the raw files GitHub archived with their SHA-256 lists, the two other manuals and `DISCLOSURES.md`.

Quick Reference: The Card for the Glove Box

Keep this card beside the screen that shows the wall. Every line is explained in the chapter named beside it.

The two sides (chapter 3). Amber is native, the system by itself. Blue is Omni-Compass on top of native. Nothing else is drawn.

The three words (the key). Green GAIN, yellow EVEN, red LOSS. A resource that rose is *power given*, green. White with a dashed edge is *the runs disagree* or *the runs differ*; grey is *shown, not judged*; neither is a reading.

The switch (chapter 5). At the top of the wall. Green ON: the governor's outputs reach the muscles. Red: the kill switch engaged, every setting handed back to native at once, the decisions shaded red. White OFF: the

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brain reads in shadow and nothing it computes leaves it; every Omni-Compass side follows native in native's colour.

The four signals (chapter 4). The work waiting, the work done, the decisions past their line, the capacity held: latency, traffic, errors and saturation in site-reliability terms; native beside Omni-Compass on each.

The alarms row (chapter 6). Every reading that went against native, worst first, each with its source record and its root-cause document where one exists. ALERT, WARN, INFO as an operations log grades them.

Where a number comes from (chapter 27). Click the tile: it opens the monitor it is a copy of. The monitor names the table, the run and the file; the file's SHA-256 was matched before it was drawn. Open the table in `results/live/`; open the raw files on the `raw` branch.

When a tile reads LOSS (chapter 28). The row stays. Read the three-run reading, not the one run. Open the root-cause document. The cause is found in this order: our wiring, native's setup, the scoring. The test runs again. Nothing is tuned toward GAIN; where Omni-Compass cannot bring a knob a gain, that knob stays native.

The console (chapter 18). Every knob acts on both sides alike. Demand ×1 is the day as written; noise σ, sensor delay, and four faults of 40 decisions each (a surge, half the machines lost, a sense blind, a security hold); the dial (the max or the tune-down); pause, one decision, 1× to 60×; the same day again or a new day; every knob to its start.

The age of the page (chapter 7). The last tile of the wall. The page is rebuilt by GitHub on every change to the engine, a model or a result, and every three hours.

What the page is not (chapter 29). It decides nothing. Class S views are models, not proof. The records of authority are the result files, the raw files and their SHA-256 lists, the manuals and `DISCLOSURES.md`.

Disclosures, Declarations and Disclaimers

This page is the one place every declaration about Omni-Compass is made. The README, the manual, the book, every guide and every report point here. Where any other page seems to say more than this page, this page governs.

Rights

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What a result is, and what it is not

1. Every result carries its evidence class: **T** a theorem, **V** verified in code, **S** our own models (modelled), **L** live software (real Kubernetes on GitHub's machines), **P** a physical meter (a real card's own power meter). A model is a statement about the model. A software benchmark is a statement about the software and machine it ran on. Only a meter speaks for hardware.
2. The models were written by the same people who wrote the law. They show how the mechanism behaves; they are not an independent test.
3. On Kubernetes in kind every machine stays powered, so energy there is a declared model, not a meter.
4. Every comparison is paired: the same seed, load and clock with and without Omni-Compass, in an order rotated by repetition. Intervals are 95% intervals over the paired repetitions. Ratios across seeds are summarised as geometric means. Labels are assigned by code from rules written and committed before the run (`docs/GPU_PREREGISTRATION.md`, `docs/REALMS_PREREGISTRATION.md`, `docs/K8S_COMPASS_PREREGISTRATION.md`), never by hand.
5. Losses and failed runs are kept beside the wins (`docs/EVIDENCE_LEDGER.md`, `results/live/`, git history). The first real-card run, for example, saved energy (+3.6% work per energy, proven) and made the slowest answers 58.5% slower; it is published as it happened.
6. No result is a promise of a saving on any other system. No dollar figure in this repository is a measured result. Nothing here is a vendor certification, a safety certification, legal advice or investment advice.

Do no harm, and the rule Omni-Compass keeps on every muscle

What "do no harm" means (the founder's words of 19 October 2026; this is its meaning everywhere in Omni-Compass, and it has no other). Omni-Compass has one goal: to strengthen the organism it governs and keep it healthy, working at the best state its own muscles can carry. Whatever is fed into the organism (its work, its data, its energy: its nutrients) passes through its muscles and comes out as its output.

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Omni-Compass's brain drives the organism toward the most and the best output it can make from what it is fed (the most work, done fastest, with the fewest misses and failures) while it keeps every muscle healthy.

- **Harm is judged at the organism.** Harm is the organism turning out less than its native controllers make it turn out (less work done; more time waited, missed or failed), or the organism made less healthy: a muscle pushed past its own limit, a service pushed over its own line, or anything made to fail. Omni-Compass does neither.
- **Every muscle only as far as it can go.** Omni-Compass can push a muscle only as far as that muscle itself can be pushed: its own limit, read live from the system, never past it. It finds the room each muscle already has and uses all of it. Whether its mathematics can draw a little more out of a muscle than the muscle was built to give is an open question; a measurement answers it before anything is claimed.
- **The muscles work as one body.** Where one muscle's strength cannot be used without straining another, the brain offsets and engages the muscles together to bring the whole organism to its best state. It may use one muscle less, leave it native, or rest it, so that another can carry more, whenever the organism's whole output comes out ahead and every muscle stays healthy. Which muscles to drive and which to leave native is the brain's live decision, at every moment and in every run.
- **Reactive and proactive.** Like an immune system, the brain answers a breach at once, and it acts before a line is crossed.
- **Every row is still shown.** Each muscle's and each gauge's reading stays in every table beside the organism's, LOSS included. Nothing is cut.

How the engine that stands keeps it. The engine that stands keeps do no harm in its strictest form. By default it moves each muscle's settings toward more work, faster, up to the muscle's own limit and never past it (the max); on the dial anyone can tune it down toward the same work with less (the tune-down). Either way it keeps a step only where the muscle is never behind native for it: a step that costs the muscle more for each piece of work, or loses more than **2%** in any measure (work done, response time at the median, p95 or p99, time over the service line, or the machine's own cost of running Omni-Compass), is undone. A resource spent that rose is **power given**, shown beside the work, the speed and the wear it bought, never called a loss. Where no move passes that test, Omni-Compass leaves the muscle exactly as it runs alone. It takes nothing from one muscle to give to another: that form never harms the organism, but it leaves unused any gain that needs one muscle to give way to another. Letting one muscle give way so that the organism's whole output comes out ahead, judged on the organism and on every muscle's health, is the next engine version's work. It is built and proven in our own models first (the toy gate), every benchmark is run again on it, and no result claims it before its runs exist.

The factory tune. The engine in this repository is presented as The Omni-Compass LLC tuned it and ships it: a self-running engine, set at the factory. Every muscle in the catalog is wired, and each runs on the setting found for it: its own card where one was found on development seeds and kept under the strict rule (364 muscles, `realms/muscle_cards_v5.csv`), the engine's own setting for every other muscle, all frozen and fingerprinted by version (`python3 tools/omni_version.py`). Muscles differ: no single setting is right for every muscle, family, sector, domain or realm, and none is presented as universal. Every published

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result is this factory tune. As with any engine, a licensee may then add on, retune, upgrade or tune down any muscle or any combination of muscles (the dial, the cards, the bands and the service lines are all exposed); a changed tune is the licensee's own configuration, and its results are its own, not the published ones. On a live system the verdict (`omnicompass/verdict.py`) measures it with paired trials on the muscle itself before every step. Every result reports each measure, and where one read LOSS at all, by however little, the result says so and by how much.

The wiring declaration

THE PUBLISHED BENCHMARKS ARE THE RECORD OF WHAT OMNI-COMPASS DID ON THE SYSTEMS, LOADS AND ENGINE THEY NAME, EACH WITH ITS EVIDENCE CLASS. WHERE A RESULT ON ANY OTHER SYSTEM DIFFERS FROM THEM, OR ANYTHING LOOKS WRONG, THE FIRST PRESUMPTION IS THE WIRING, THE SETTINGS, OR HOW OMNI-COMPASS WAS IMPLEMENTED OR READ, NOT THE LAW. CHECK THOSE FIRST (THE MANUAL'S "WIRED RIGHT OR WIRED WRONG") BEFORE DRAWING ANY CONCLUSION ABOUT OMNI-COMPASS.

Omni-Compass acts only through the wires it is given. It reads the meters it is pointed at and moves the settings it is allowed to move. If a reading is the wrong one, slow, or blind; if a lever is the wrong one, already owned by another controller, or given a range (cover) that is wrong for the machine; if a band or a service line is set for a different workload; then Omni-Compass will do exactly what its law says with the wrong information, and the result will not be the published one.

Therefore:

1. **It cannot be slapped on.** Every installation goes through the levels of the manual in order (chapter 6), starting read-only, and every writing level begins with the wire check (manual, section 4.4) and a watch arm that must equal native.
2. **If your paired receipts differ from the published benchmarks in direction** (service LOSS, or no saving where one was shown), **the first presumption is wiring, not the law.** Confirm the installation with the checks in the manual's section 4.5 (*Wired right or wired wrong*) before drawing any conclusion about Omni-Compass.
3. **This has happened to us, and is published.** On the first real card (NVIDIA A10, 2026-10-02) the governor read being busy as trouble, set its power lid under the card's own working draw and allowed the clock under the card's own working clock; the card served bursts more slowly and the slowest answers were 58.5% slower. The cause was found in the card's own samples and corrected (`docs/GPU_PREREGISTRATION.md`, amendments 6 and 7). That is what a wiring fault looks like, and how it is found.
4. Omni-Compass never writes outside a lever's cover, never fights another writer (it stops and leaves that lever alone), reads back every write, and returns every lever to the value it read before its first write when

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it stops or when the OFF switch is used. These protect the machine; they do not make a wrong wiring right.

Safety and responsibility

1. Run watch mode first; run the wire check before any write; keep the master OFF switch (`python3 tools/omni_switch.py off`, the whole harness at once) in reach at every level, and use it at the first sign of anything wrong, a suspected breach included.
2. Omni-Compass is a supervisory governor. The machines' own controls (firmware, autoscalers, safety systems) stay in place and keep their own protections; Omni-Compass sets only values they already accept.
3. Do not connect Omni-Compass to safety-critical systems (vehicles, medical devices, grid protection, life safety) except as a modelled study, and never without the qualified review those systems require.
4. The software is provided "as is", without warranty of any kind, as stated in `LICENSE`. The operator is responsible for its use on their systems.

Changes to the work and to this page

1. The software, the law's settings, the benchmarks, the results, the documents, the licensing terms and this page may change at any time, without notice. The version in the repository's `main` branch at a given commit is the version that commit describes; `RELEASE_MANIFEST.json` and `python3 verify.py` identify it.
2. Statements about work in progress, planned features, expected results and future runs are expectations, not promises. Where a result has not yet been measured, the documents say so (`docs/STATE_OF_PLAY.md`, `docs/DOSSIER.md`, section 8).
3. A signed Omni-Compass Enterprise License governs its own terms for its own term.

Privacy and data

Omni-Compass contains no telemetry. It sends nothing to The Omni-Compass LLC or anyone else, and makes no network connection of its own except to the systems an operator points it at. Its logs and receipts stay on the operator's machines.

Where everything is

Question	Page
What has been measured, and how strongly	`docs/STATE_OF_PLAY.md`, `docs/DOSSIER.md`, `docs/EVIDENCE_LEDGER.md`

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Question	Page
The rules written before each run	`docs/GPU_PREREGISTRATION.md`, `docs/REALMS_PREREGISTRATION.md`, `docs/K8S_COMPASS_PREREGISTRATION.md`
How to wire it, level by level, and how to confirm it is wired right	`docs/OMNI_COMPASS_MANUAL.md` (chapters 4 and 6), the book `docs/OMNI_COMPASS_MANUAL.pdf`
The license, in plain answers, and third-party components	`LICENSE`, `LICENSING_FAQ.md`, `ENTERPRISE_LICENSE.md`, `NOTICE`, `THIRD_PARTY_NOTICES.md`, `PATENTS.md`, `TRADEMARKS.md`, `SECURITY.md`, `.github/CLA.md`
The muscles, what each is for and how it is wired	`docs/MUSCLE_CATALOG.md`
Proof the code is the code that ran	`python3 verify.py`, `RELEASE_MANIFEST.json`, `results/SEAL.json`

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PART ONE

The Page as a Monitor

What the live engine page is: a monitor of the engine and of the runs GitHub ran, never a drawing; how it is built, its table of contents, and its two sides and three words.

CHAPTER 1

A Monitor, Not a Drawing

1.6 What the page is

The live engine is a monitor of the Omni-Compass engine and of the benchmark runs GitHub ran. A monitor shows what its instrument reads; a drawing shows what its author believes. The page is the former by construction: every record it replays is made again by the repository's own code on the checkout it is built from, and the build refuses to write a page whose engine is not the one `python3 tools/omni_version.py` reads. So the page can never run ahead of the repository, and a new engine version is written into the page's words by a person before the page can be rebuilt on it.

1.7 The five records

The page carries five kinds of record, and every monitor says which it plays.

Record	Made by	What it holds	Class
The harness	`docs/engine/export_harness_trace.py`	the realms harness (`realms/harness.py`, the max of `realms/compass_arm_v5.py` with the bone, the twins and the immune system): one organism of sixty muscles, and each muscle alone, native and with Omni-Compass on top, decision by decision	S

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Record	Made by	What it holds	Class
The controller	`docs/engine/export_controller_trace.py`	the shipped Kubernetes governor (`omni_controller/controller.py`) on the toy cluster, on top of the autoscaler at the max and in the cloud's setting, every tick and every audited decision, runs A, B and C of six hours	S
The runs	`docs/engine/export_runs.py`	the official runs as recorded: every three-run table's runs A, B and C read back from their raw files, every file's SHA-256 matched against the run's own list, every Kubernetes repetition scored by `tools/live_reps.py` and checked against the run's result	L
The scenarios	`docs/engine/export_scenarios.py`	the GPU card (`realms/gpu_card.py`), the Linux host (the conveyance of `tools/toy_gate.py`), every muscle and organism of the realms table (`results/realms_v5`), and every judged row of the newest engine's receipts (`results/live/V13_*.md` as they land)	S, with the receipts' own classes
The build	`docs/engine/build_live_engine.py`	the engine, its digest and the commit; the time; GitHub's runs in progress and queued; every run started for the chips and the clouds (`docs/engine/github_ledger.json`)	the record of the build

1.8 What runs live

Three parts of the engine run in the page itself: the brain (`omnicompass/core.py`), the compass law (`omnicompass/compass_law.py`) and the verdict (`omnicompass/verdict.py`), each ported to the browser and proven equal to the Python and the C++ on every build by `docs/engine/check_live_engine.py`. They drive the stress-test model of sixty muscles (the page's sections 5a, 6, 7a to 7c) and the page's own render loop (the page's section 10). Nothing else computes on the page: every other monitor plays a record.

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1.9 What the page decides

Nothing. Every word on it is the reading its record carries under the one key (`tools/readings.py`), applied once, in the repository, when the table was written. The page never reads a word off the stream it draws. When a monitor has no record yet (a cloud set still running, a chip run not yet started), it says what the repository holds and shows nothing in its place: the empty slot is itself a true reading of the record.

1.10 The appraisal

Every view of the page is reproduced from the repository's own code by an independent path and compared value by value (`docs/engine/APPRAISAL.md`; rerun with `docs/engine/check_live_engine.py` and `docs/engine/check_records.py`). The brain's mathematics agree to machine precision; the harness views and the live controller are identical to the repository's own runs on every tick and every plant. Chapter 31 says how to rerun the appraisal on your copy.

CHAPTER 2

The Engine Table of Contents and the Numbering

2.5 The page's own contents

The page opens with its legal notice (i), then its own table of contents (ii, "Engine table of contents"), the title page with the build (iii) and the executive summary with the reading key (iv). The body is numbered 1 to 10; the back matter is v (disclosures, governing law and contact) and vi (the closing legal notice). A section's number is printed before its title; a lettered section (1a, 4c) is a monitor inside the numbered part that holds it.

2.6 The sections, as built on omni-v13

Appendix A lists every section with its anchor and the chapter of this book that reads it. In the page's order: 1 the monitor wall (1a the status board and the wall, 1b the pipeline); 2 the whole organism; 3 the measured effect by domain; 4 the evidence by stack (4a the clouds, 4b the orchestrators and governors, 4c Kubernetes, the official run playback, 4d Kubernetes, per-repetition telemetry, 4e the shipped controller's decision trace, 4f the chip and the host); 5 the organism live (5a native and Omni-Compass, 5b the telemetry analytics and Omni-Compass in its own words); 6 the console; 7 the brain (7a one cycle, 7b the phase plane, 7c the brain's state, 7d the verdict); 8 the failure modes; 9 the component specification; 10 the self-governed rendering.

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2.7 Anchors

Every section has an anchor that does not move when the page is reordered: `#p-health`, `#s-health`, `#s-github`, `#s-whole`, `#s-fleet`, `#p-evidence`, `#s-cloud`, `#s-orch`, `#s-ops`, `#s-runs`, `#s-live`, `#s-stacks`, `#p-organism`, `#s-engines`, `#s-lab`, `#s-desk`, `#p-brain`, `#s-decision`, `#s-plane`, `#s-brain`, `#s-verdict`, `#s-fail`, `#s-math`, `#s-self`, `#s-disclose`. A tile on the wall links to the anchor of the monitor it copies; a row of the table of contents links the same way. When this book names a section it gives its title and its anchor, so the reader finds it on a page whose numbering has moved.

2.8 The order, and why

The founder's order of 20 October: the wall first, then the benefit board, then the whole organism, then the evidence from the cloud down to the chip, the organism live, the console and the brain. The order is the order a person reads a system: is it healthy now (the wall), what has it bought (the board and the whole), where is the proof (the evidence), what is it doing this second (the organism live), what can I do to it (the console), and how does it think (the brain). The disclosures close the page as they open it.

CHAPTER 3

The Two Sides and the Colours

3.5 The sides

Native is amber, Omni-Compass is blue, everywhere: every line, bar, tile and number that belongs to one side wears that side's colour, in every chart and table. Where a monitor draws the two sides on one axis, native is the left lane or the upper row; where it draws them as two instances (the organism, the phase plane), native is on the left.

3.6 The words

GAIN is green, EVEN is yellow, LOSS is red. Every reading, in every table, chart, bar and tile, wears one of those three colours and no other. Power given wears GAIN's green. *The runs disagree* and *the runs differ* are white with a dashed edge, and *shown, not judged* is grey, because none of them is a reading. The size printed beside a word is how far the gauge moved from native and carries the word's own sign (GAIN +, LOSS −, EVEN =), never the gauge's raw direction. A change under half of one percent either way reads EVEN, under the floor.

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3.7 The sides are never a reading

Blue does not mean good and amber does not mean bad: they name who is being read. Where the two sides look the same, they are the same, and that is EVEN. Where Omni-Compass's blue line sits closer to a muscle's line than native's amber one, that is the max at work: every muscle to its own limit, never past it, and the capacity it holds there is power given, shown beside the work and the speed it bought.

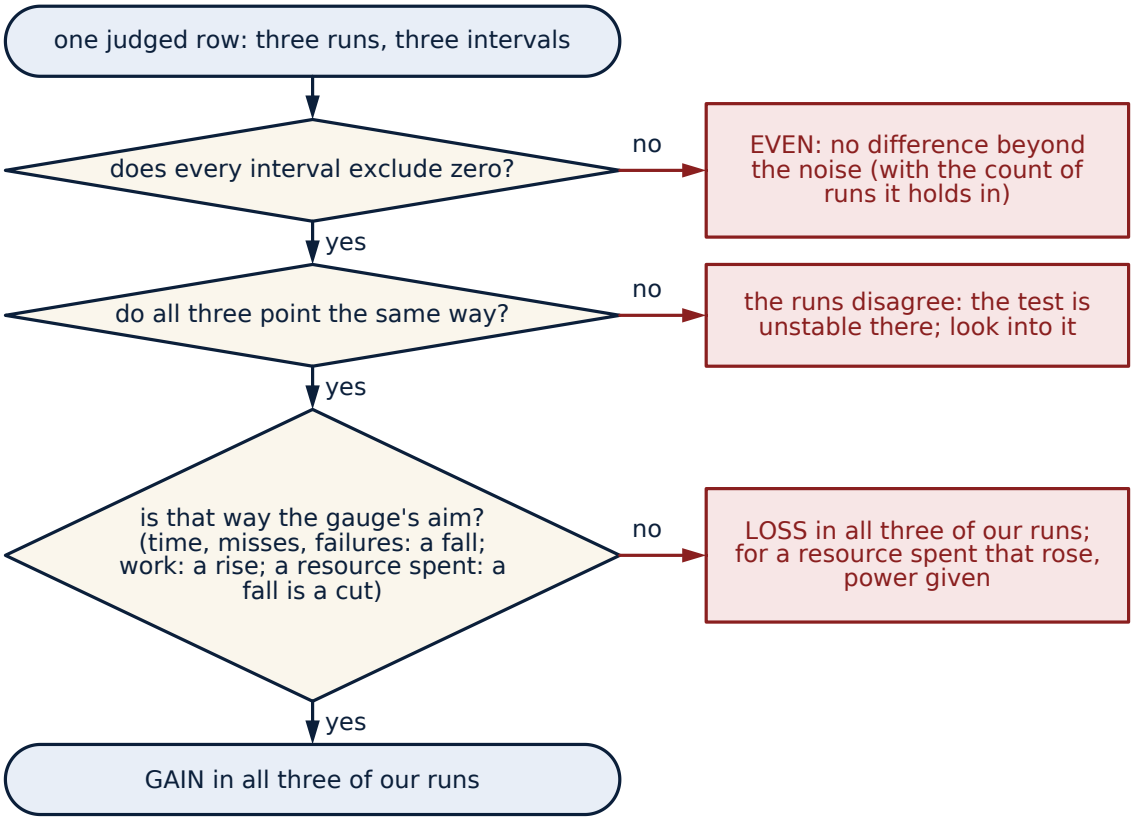


Figure 1. Flowchart: how a gauge's number becomes a word. The gauge's declared aim is read first: time waited, missed or failed, where a fall is GAIN; work done, where a rise is GAIN; a resource spent or held, where a fall is a cut (GAIN) and a rise is power given. Then the paired change and its 95% interval: an interval that includes zero reads EVEN; clear of zero, the word follows the aim.

Three runs A, B and C: GAIN or LOSS only when all three point the same way with every interval clear of zero; EVEN when any interval includes zero; the runs disagree when clear runs point different ways. The page draws exactly this rule, as tools/readings.py writes it.

3.8 The marks

A kill-switch window is shaded red across every time axis. A monitor on screen now carries an accent border on its tile. A tile with no reading says what it shows (a count, a state, an age) and carries no colour of a word.

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A row shown and not judged (the knob's own setting, the governor's own commands, Omni-Compass's own processor time) is grey and counted in no total.

PART TWO

The Wall and the Switch

The status board, the master switch and its three states, the alarms, the golden-signal feed, the event log, the metrics exposition, and the pipeline that rebuilds the page.

CHAPTER 4

The Monitor Wall

4.6 What the wall is

The status board (the page's section 1a, `#s-health`) is every live monitor of the page in one view: the same canvases the sections draw, copied small four times a second, so nothing is drawn twice from different data. Click a tile and the page scrolls to the monitor it copies. The page's own governor (the page's section 10, chapter 25) leaves a monitor resting while it is off screen; while the wall is in view the resting monitors are drawn in turn, one a frame, and a tile with the accent border is a monitor on screen now.

4.7 The order of the board

The board is laid out the way an operations monitor is. The switch and the engine come first. Next come the live organism's signals, native beside Omni-Compass, in the four a service is watched by (latency, traffic, errors and saturation in site-reliability practice): here the work waiting, the work done, the decisions past their line and the capacity held. Then every reading of the real and modelled runs that went against native, worst first. Last come GitHub (the runs in progress and queued) and the age of this page. A tile's word is its reading against native by the key; a tile with no reading says what it shows.

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4.8 Reading a signal tile

Each signal tile holds the last 120 decisions of both sides, native solid amber and Omni-Compass blue, each side's value now, and the running paired reading in the tables' words. With the master switch OFF the decisions are shaded red and Omni-Compass's side follows native: its value equals native's and its line is drawn in native's colour, dashed on top of native's, so both stay in view.

Signal	What it is	Its aim	A GAIN reads
The work waiting	the queue in front of the muscles: requests, jobs or pods not yet served	time waited: lower	the blue line under the amber one
The work done	the requests served, the jobs finished, the work inside the line	work done: higher	the blue line over the amber one
The decisions past their line	the share of decisions where a muscle's reading stood past its line (a miss)	missed: lower	the blue line under the amber one
The capacity held	the machines, replicas or clock a side holds in service	a resource held: a fall is a cut, a rise is power given	either way is shown beside the three above; a rise that bought nothing is a cost to find

4.9 Reading an alarm tile

An alarm tile is a judged row of a real or modelled run that read LOSS, or a run on the engine that stands that terminated without a measurement. It names its gauge, its size with the word's sign, the table or run it came from, and the repository's root-cause document where one exists (`docs/POD_START_WAIT.md`, `docs/TAIL_LATENCY.md`, and the others). The alarms are ordered worst first by their size. Chapter 28 is what to do with one.

4.10 The age of the page

The last tile. It counts from the build's time, printed in the title block, to now on the viewer's clock. A page older than three hours and a few minutes means the rebuild did not run or did not land (chapter 7); the records on it are still true for the build they name.

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CHAPTER 5

The Master Switch

5.5 One switch for the whole organism

The master switch (the page's section 1a, under the wall) is wired at the top of the organism: switch, brain, organism, every muscle, as the repository builds it (`omnicompass/master.py`, `tools/omni_switch.py`). It holds the brain's outputs. Every governor on the page is registered with it and obeys it at once; on a real machine every governor exits at the switch.

5.6 The three states and their colours

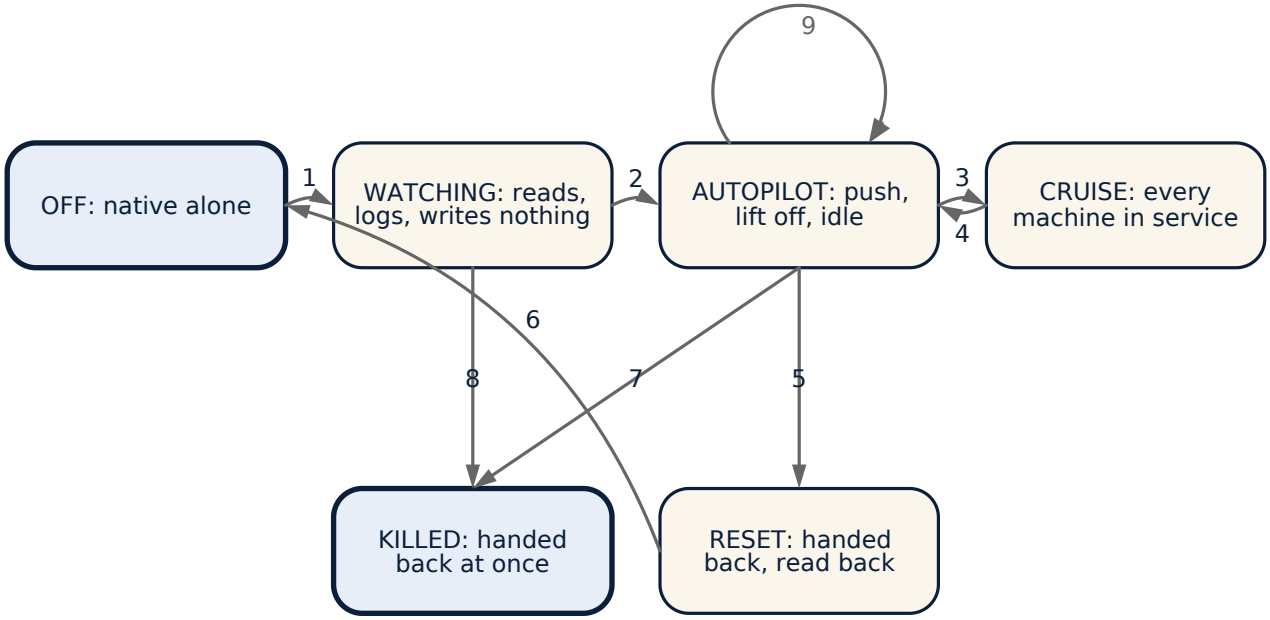
State	Colour	What it means on the page	What it means on a real machine
ON	green	the brain's outputs reach every muscle; every monitor shows Omni-Compass acting on top of native	the governors run; every write read back; the lease renewed each beat
Kill switch engaged	red	every lever handed to native at once; the decisions shaded red on every time axis; the event log records the engagement and its clearing	`python3 tools/omni_switch.py off`: every governor hands back and exits, and none starts again until `on`
OFF	white	the brain keeps reading the native body in shadow, so its monitors keep moving, and nothing it computes leaves it; every Omni-Compass side follows native, drawn in native's colour, dashed	the governor in watch mode writes nothing (the Operator's Manual, level 1); off, it is not running

No monitor is switched off by the switch. That is deliberate: the operator sees what the brain would do while it does nothing, which is the watch level of every installation.

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5.7 The table under the switch

Governor on this page; where it lives (the module, the section); its state; what it does at the switch. Every governor the page runs (the stress-test model's, the harness's, the live controller's replay, the page's own render governor) is a row, so there is no governor on the page that the switch does not reach.



1 level 1 · 2 the wire check passed; level 2 and up · 3 pods waited 2 decisions · 4 the line empty 2 decisions · 5 the reset file: the end of a run · 6 exit 0 · 7 the kill switch, from any mode · 8 the kill switch · 9 idle at once when demand is at zero

Figure 2. State machine: the governor's modes. OFF is native alone; WATCHING reads and logs and writes nothing; AUTOPILOT pushes, lifts off and idles; CRUISE holds every machine in service while a known queue waits; RESET hands every setting back and reads it back. The kill switch leaves every mode for OFF at once; the lease walks a crashed governor's settings back to native. On the page the switch's three states are ON (green), the kill switch engaged (red) and OFF (white, the brain reading in shadow).

5.8 What a kill-switch window looks like

Press the kill switch on the console (chapter 18) and every time axis on the page shades the window red from the press to its clearing. Inside it, Omni-Compass's lines fall onto native's; the governor's ticker (the page's section 4c) is empty, for nothing decides on top of native; the phase plane's right-hand dots take the left-hand positions. When the window clears, the lines part again where the governor's first decision lands.

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CHAPTER 6

The Alarms, the Feed, the Event Log and the Metrics

6.5 Alarms and watch items, by severity

The alarms table under the wall lists every reading that went against native and every open item (a run that ended without a measurement, a slot not yet run), graded ALERT, WARN and INFO as an operations log grades them, each with its item, its reading, its source and the repository's record. Filter by the three grades with the buttons at its head. An ALERT is a judged LOSS in a three-run table on the engine that stands; a WARN is a LOSS in one run of a table not yet complete, or a run that stopped before a result; INFO is a watch item.

6.6 The golden-signal feed

One point per decision streamed from the live engine, gapless, the last 240 decisions, each signal on its own axis, with each side's last, mean and maximum as a time-series dashboard reports them; a kill-switch window shaded red; native in amber, Omni-Compass in blue, the same demand on both. The feed is the four signals of the wall at full size and full history.

6.7 The event log

The live engine's own events as it emits them, newest first, with the wall-clock time and the decision: a threshold breach and its clearing on either side, every injected fault and its end, the kill switch, the dial, a new day, and the brain's calm crossing the grant threshold. The log is what an operator reads to answer "what happened at 02:14": the decision number ties it to every other monitor.

6.8 The OpenMetrics exposition

The live engine's metrics in the OpenMetrics text format, the exposition format Prometheus scrapes and Grafana charts, refreshed every two seconds. They are: engagement (the switch), the decision index, breaches, work, band time, actuator moves and reversals per side, and frame time; then the verdict's keep ratio at each true cost (chapter 22), the three-run tables' rows by class and reading, and the build's engine, digest and

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commit. *Copy the exposition* puts the text on the clipboard, so the page's own metrics can be pasted into a scrape target and charted beside the stack's.

CHAPTER 7

The Pipeline That Rebuilds the Page

7.4 When the page is rebuilt

GitHub rebuilds the page (`.github/workflows/live-engine.yml`, `docs/engine/build_live_engine.py`) whenever the engine, a model or a result changes, and every three hours. Every record is made again by the repository's own code on the engine that stands, checked against the Python and the C++, and committed. The page's own file is outside its own push filter, so a rebuild never starts another rebuild; a front-page run that did not succeed starts no rebuild (`tests/test_workflow_triggers.py` holds both).

7.5 The CI pipeline monitor (the page's section 1b, `#s-github`)

Every workflow run in progress or queued at the build, as GitHub's own list at that moment, and the first of them in each state. The hardware and cloud runs, with their terminal status, are in the page's sections 4a and 4f; every run on the engine that stands that terminated without a measurement is raised on the status board.

7.6 Reading the pipeline against the conductor

The conductor (`scripts/dispatch_grid.py`, every 15 minutes) starts every run of the campaign on the engine that stands, A of every benchmark, then B, then C, a few a tick, and never starts one twice; it starts nothing while an Azure run, a grid report or a verify waits for a runner. So a pipeline monitor showing one Azure run queued and nothing else starting is the conductor holding, by its rule, not a fault. A long run on a free runner (the organisms at 1,000 copies) lists as in progress for its whole shard.

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PART THREE

The Whole and the Measured Effect

The whole organism two ways, its bills and receipts, the benefit board, the grid and the catalog.

CHAPTER 8

The Whole Organism, Two Ways

8.7 Two ways a stack is bought

Omni-Compass is wired into every piece of a stack through its own two-way plug: the software (the cluster, its add-ons, its governors and orchestrators, and the services on it: the pools, the consumers, the caches), the bare metal under it (the chip's power limit and clock; the machine's kernel and its own meter) and, where the stack is rented, the cloud that bills the machines. A stack is bought one of two ways, and the whole-organism monitor (the page's section 2, `#s-whole`) has a button for each: *on-site bare metal*, the software on the owner's own bare metal (software × bare metal, no cloud); and *through the cloud*, the cloud's build, its bare metal at the back end and the software (cloud × bare metal × software), with the cloud as two pieces beside it. The founder's question of 20 October, which this monitor answers: the cloud is a piece of the stack, bought and billed, and what Omni-Compass does there composes with what it does on every piece beneath it.

8.8 The stack, drawn

The stack, native beside Omni-Compass, with the work flowing down through the pieces: the service, the cluster, the governors, the machines, the chip, and, through the cloud, the provider's bill. The dots move at the live engine's demand; the picture is of the flow, not a measurement; each piece's measured factor replaces the picture's weight the moment its three runs land.

8.9 The collective synopsis

The benefits, gauge by gauge, added together over every measured piece: work (more), speed (a shorter 95th percentile), machines (fewer), energy (less), the bill (smaller), and the one number, the index over the gauges. Two readings are offered by the buttons: *every measured piece, every interval carried* (every factor with its own 95% interval, the whole their product) and *the strict reading: the index's rule* (only a factor that reads GAIN or power given in all three of its runs counts; EVEN counts as exactly 1). A third pair of buttons says

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what to do with a governor that replaces a native controller: *its one run beside the count* or *its one run in the count*.

8.10 The composition

Piece by piece and layer by layer: each factor with its 95% interval, the whole their product. Read the table left to right: the piece and its layer, the five gauges, the index, and the receipt (the table's own row) each factor is read from. A factor's interval that includes 1 is EVEN and multiplies the whole by exactly 1 in the strict reading.

8.11 Bills and receipts

What native spent (the cost; runs A, B and C), what Omni-Compass spent (the expense; A, B and C), and the benefit (pooled, with its 95% interval), in the units each receipt bills in: machine-hours, dollars at list price, watt-hours of the declared model, the meter's watt-hours where a meter exists. The cost is native's, the expense is Omni-Compass's, and the benefit is their difference read by the key: a smaller bill is GAIN, a larger one is power given, shown beside the work it bought.

8.12 Beyond the count

The governors' tables on this page, the models beside them, the separability test (whether the pieces' factors may be multiplied: the factors are independent where each piece's run holds every other piece at native), and where the missing pieces are: every slot not yet measured is named, and the whole is read without it rather than with a guess in its place.

CHAPTER 9

The Benefit Board, the Grid and the Catalog

9.9 The benefit board (the page's section 3, `#s-fleet`)

Every domain, native against Omni-Compass, as the repository's benefit sheet holds it (`results/BENEFIT_SHEET.json`, `docs/BENEFIT_SHEET.md`): the domain, the benchmark, its evidence class, its engine, its benefit (one number and one word), its GAIN rows, power given, EVEN and LOSS counts, and where it stands (landed, running, queued, not yet run). Each bar and square carries its own word; the size beside the word never carries a sign of its own.

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9.10 Where the benefit lives, gauge by gauge

Every judged row of every newest three-run table, sorted into the family of gauge it measures. A CTO reads speed, the service level, failures and work. A CFO reads machines, energy and the bill. A plant engineer reads peaks and swings, wear and accuracy, and misses. Each bar counts the rows that read GAIN in all three runs; the size beside it is the median of those rows' own sizes; the largest is printed beside the median; power given and LOSS rows are counted beside them, and the runs that disagree or differ are listed, not counted.

9.11 Aggregate: one figure for the whole board

The Omni index (`results/OMNI_INDEX.md`): the geometric mean of the ratios of every measure that enters, where a measure enters only as its three-run reading allows (GAIN, LOSS or power given in all three runs counts; EVEN counts as exactly 1, so nothing inside the noise is claimed either way). Each test is read from its own newest three-run table, no table is pooled across engines, and every row names its table. The index is one number for the board; its row of factors under it is the board's own audit.

9.12 Third-party software still to be benchmarked

The register's slots, in order (`docs/REGISTER.md`): every muscle in them already runs, modelled, in the 1,107-muscle catalog and the seven organisms, native and with Omni-Compass on top; a slot is the same test on the outside world's own software, each queued until its three runs land. A slot shows its status in the register's own words, what is measured (the lines of its preregistration) and the known limits said before the result.

9.13 The scale-out grid

Seven organisms × 1, 10, 100 and 1,000 copies × 1, 10, 100 and 1,000 runs: modelled (the misses) and with the real cluster inside (the 95th percentile), both values shown in every square. The 1,000 tier runs on GitHub's free runners only; paid machines stay at 100. Two buttons choose *the engine that stands: its runs as they land* or *the road: the newest grid with its squares landed*. A square not yet landed says so; a square beyond a free runner's memory or time is listed as pending runner capacity, never rented.

9.14 Every three-run table

Every three-run table in the repository, native against Omni-Compass, every row as written, each benchmark's newest engine's table; open a row to read every judged row that is not EVEN. The table's own rule is printed with it; rows shown and not judged are left out of the counts; a tuning case is shown and never counted.

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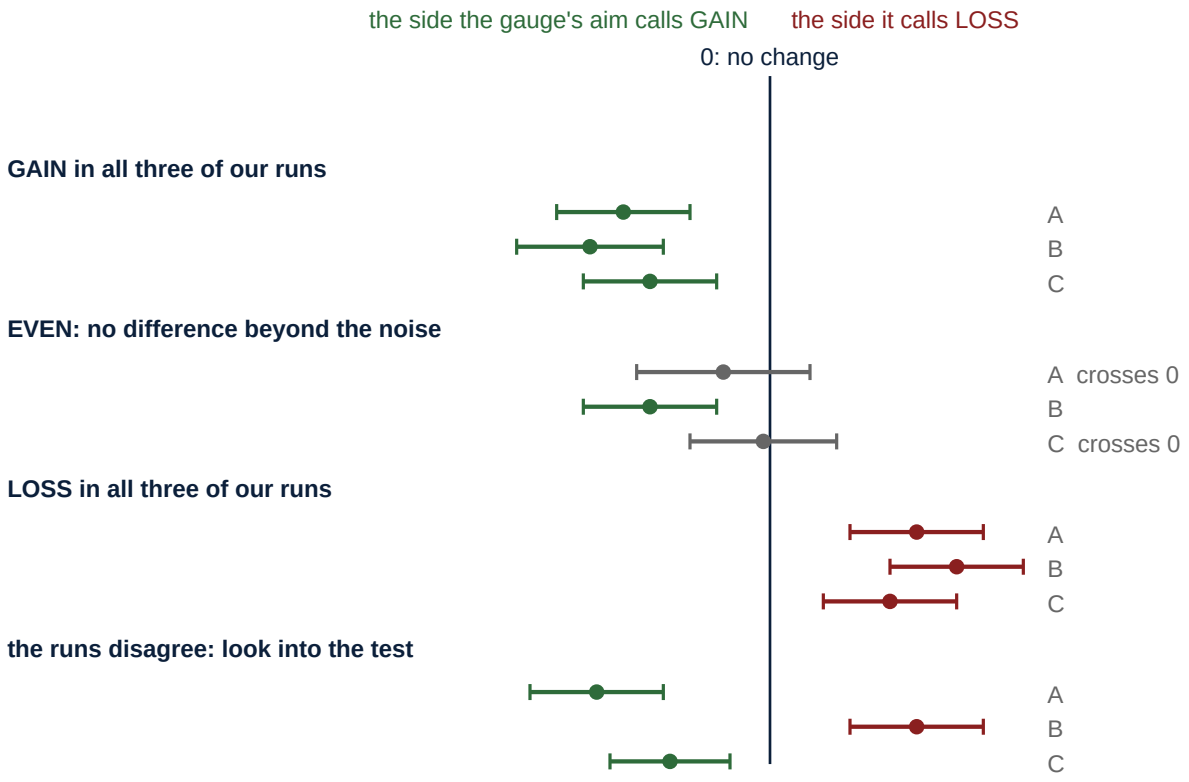


Figure 3. The three-run rule. A benchmark runs three times, A, B and C, on one engine version, each a paired run of native against Omni-Compass. A row reads GAIN or LOSS in all three of our runs only when the three agree in sign with every 95% interval clear of zero; EVEN, no difference beyond the noise, when an interval includes zero; the runs disagree when clear runs point different ways. The board and the index count by this rule and no other.

9.15 Every muscle and the seven organisms

From `results/realms_v5/REALMS_V5.json`, the realms runner at the max, its engine and commit named above it (`tools/run_realms_v5.py`, catalog `realms/catalog_v5.csv`), ten paired seeds per muscle and per organism, the lease after 80% of every run. Misses: the plant decisions over their line, summed over the ten seeds. A muscle reads GAIN when one of its aims (more work, faster, fewer misses) moved beyond the noise under some load with none moving the other way; EVEN at its limit on native when its knob already sits at the top of its range; EVEN, the same as native, when it wrote and nothing moved beyond the noise. Energy is power given where it rose. The seven organisms are listed spine first, then the four realms, the four stacked and the whole tower, each under its own load, the climb and the work waiting in line (the three buttons). Evidence class S: a model's energy is not a meter's.

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9.16 Every family, by stack

Each muscle alone, native and then with Omni-Compass on top, grouped by the family of its knob and the stack it lives on; *Find a muscle* searches the 1,107 by name. A family reads GAIN when more of its rows read GAIN than LOSS; the row of a muscle at its limit on native is counted as EVEN.

PART FOUR

Evidence by Stack

From the cloud down to the chip: the clouds, the orchestrators, the official Kubernetes runs and their playback, the shipped controller's decision trace, the card and the host.

CHAPTER 10

The Clouds

10.6 One metric: the bill

In a cloud Omni-Compass shows one thing: the same work as native for a smaller bill, never a bigger one. Every cloud set runs Omni-Compass at the tune-down on the machines alone (`OMNI_MACHINES_ONLY=1`, `--give-back-only`): no pod autoscaler moved, no processor conveyed, no power capped, so the pods are exactly native's. It parks only a machine the cloud already made, after its own trial, and wakes it on a breach. The bill and the machines aim lower; work, speed and failures must read EVEN or GAIN. Azure runs first and is the measure of the other two (Google Cloud, Amazon Web Services): the same tuning, mechanism and harness.

10.7 The monitor (the page's section 4a, `#s-cloud`)

Each provider's four sets (the steady set, the bursts `1 4 1 6 1 9` and `1 6 1 8 1 6`, the real day of Google 2011 on 1 to 9) and their status: measured to the end (engine, run, started), stopped before a result, or now. Then every cloud run GitHub has started, set by set; the bill on Azure run by run (machine-hours billed native · Omni-Compass; the bill; the 95th percentile; the failed requests), each with the engine it ran on; and the shipped controller in the cloud setting, live: native's machines against Omni-Compass's on the repository's

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toy cluster (class S), the cloud's own autoscaler running in both arms, its run and speed following the console's clock.

10.8 The stopping rule

From 19 October every Azure set runs up to 100 paired repetitions a run, the paid cap, under the preregistered stopping rule (`tools/stop_rule.py`). After 10, 20, 40 and 70 repetitions the bill's change is read with its 99% interval. The run stops once the bill is read (GAIN, power given, or EVEN within ±2% of native's bill), and never while the speed or the failed requests read a LOSS. Those runs carry "to the cap, stopping rule" in their title and count as their own A, B and C. So a cloud set that has run for a day and a half and shows no result on the board is not a fault: its bill has not yet been read at a look, and `peek` (`.github/workflows/peek.yml`) prints its finished repetitions, paired, while it runs.

10.9 Reading a cloud receipt

The rows an operator reads first: machines billed (machine-hours) and the compute bill at list price, both aiming lower; then the response time's mean, 95th and 99th percentiles, the time over the line, the failed requests and the alert minutes, all aiming lower and all required EVEN or GAIN; then the rows shown and not judged: the settings commands the governor wrote on its own (native is 0), the share of its additions decided inside the line, Omni-Compass's own processor time. A bill row whose interval includes zero is EVEN so far and the run goes on to its next look.

10.10 When a cloud is not ready

The conductor starts a cloud's set only after that cloud's newest check passed (`.github/workflows/cloud-check.yml`, `tools/cloud_key_check.py`: the key whole, the login, the plan, the machine type, every quota), at no cost. The monitor shows a failed check as the cloud's status, in the check's own words: a Google key stored cut short (the stored secret shorter than a whole key file) is stored again whole (`docs/GOOGLE_SETUP.md`, the page's section 4); an Amazon account on the Free plan, whose organization policy denies every machine read, is moved to the Paid plan (`docs/AMAZON_SETUP.md`, the page's section 1a). Nothing is built and nothing is billed until the check passes.

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CHAPTER 11

The Orchestrators and Governors

11.5 What they are

The governors and orchestrators that sit between the work and the cluster: batch schedulers and job queues (Kueue, Volcano, the KAI scheduler, Armada), multi-cluster governors (Karmada, Open Cluster Management), the default Kubernetes scheduler and its bin-packing profile, the co-location and rightsizing controllers (Koordinator, Crane, CruiseKube, Goldilocks), the GPU and edge orchestrators (KubeRay, KubeEdge, Nephio) and Gödel. Each runs live on real Kubernetes as its project ships it, native and with Omni-Compass on top of it; Omni-Compass never replaces the orchestrator.

11.6 The monitor (the page's section 4b, `#s-orch`)

The road to a result, every governor at once: its register row, its preregistration, its rig on GitHub, runs A, B and C, and its three-run table as it lands. The orchestration stack, live: Omni-Compass on top, the governor native as shipped, Kubernetes and its nodes beneath, the work dots moving at the live engine's demand (a picture of the flow, not a measurement, replaced by the governor's readings the moment its three runs land). The catalog of governors: what each governs, native as shipped, Omni-Compass on top, and what the repository holds today. Click a governor (a tile, a row of the road or of the catalog) to hold it in every view.

11.7 The four ways

The founder's order of 20 October: each governor runs with the organism alone (nativebare), the orchestrator as shipped (native), Omni-Compass on top of the organism alone (compassbare) and Omni-Compass on top of the orchestrator (compass), all four in every repetition, each comparison paired within it and read by the one key (`docs/BATCH_QUEUE_PREREGISTRATION.md`, amendment 6). On the organism alone Omni-Compass sits on top of Kubernetes' own scheduler. The monitor shows the four as four columns of one table per governor; the comparisons an operator reads first are native against compass (what Omni-Compass adds on top of the orchestrator) and nativebare against compassbare (what it adds with no orchestrator at all).

11.8 The orchestra

The eighth organism: every governor's catalog muscles, each governor's a body of its own and all of them together one body, native and with Omni-Compass on top (modelled, class S; `tools/run_orchestra.py`, `results/orchestra/ORCHESTRA.md`). It is a grouping of catalog rows like the seven, never an engine file.

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CHAPTER 12

Kubernetes: the Official Run Playback

12.8 What plays (the page's section 4c, `#s-ops`)

A real-time operations dashboard over the benchmark runs themselves (class L), not a model. Every series streams half minute by half minute out of the raw files GitHub archived, each file's SHA-256 matched before a byte is drawn: native (the whole cluster with everything it ships with) against Omni-Compass, the same load, the same second. Pick a benchmark, a run (A, B or C) and a repetition at the head of the section; the tiles, the candles, the heatmap, the bars and the two tickers all move together with the page's section 4d.

12.9 The heart

A supervisory governor is only alive while its loop turns: once every decision it reads the whole body, decides, writes, and proves it is still there. That turn is the heart. Each beat pumps every reading in through the plug and the decision out (`omni_controller/controller.py`, its main loop) and renews the governor's lease (`master.heartbeat()`, `omnicompass/master.py`). If the heart stops, its process gone or its lease older than its limit (the larger of 120 s and five beats), the watchdog (`tools/omni_switch.py watchdog`) walks every setting back to native: a stopped heart leaves the body exactly as native runs it. Native has no heart of its own: each of its controllers keeps its own clock, the pod autoscaler one sync every 15 s, and nothing proves any of them is still there.

12.10 A healthy heart, never pushed

The beat runs at a fixed, calm rate (one decision a minute in these runs, the fast checks every 5 s between beats) and is never driven faster by the load it governs. What it costs is counted inside the body's own load (the row *Omni's own CPU*), so the organism never pays for its brain without knowing it. A decision that fails is recorded and skipped: it writes nothing, the body keeps the last settings that landed, and the next beat comes at its normal time. Two gauges read its health: the share of each beat spent deciding, and how close the lease came to its limit. The page calls a heart under strain when a decision takes more than half its beat, or a gap between beats runs past half the lease.

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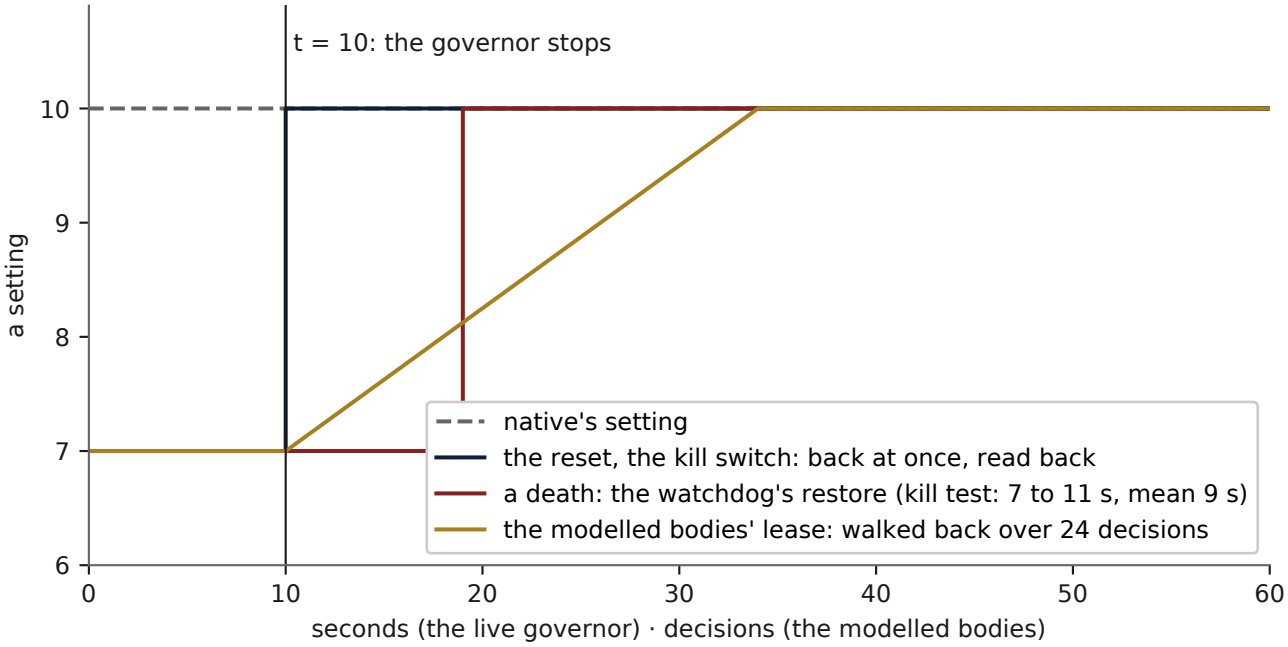


Figure 4. Timing diagram: the lease. The governor renews its lease on every beat; the watchdog reads the lease on its own clock. When the governor dies without handing back, the lease ages past its limit (the larger of 120 s and five beats) and the watchdog walks every setting back to native, a step at a time, reading each back. On the page, the heart monitor shows the beat and how close the lease came to its limit; a stopped heart leaves the body exactly as native runs it.

12.11 The three charts

Response time, one candlestick per 30-second bin (open, high, low, close of the responses in the bin), native beside Omni-Compass; the latency heatmap, every response binned by time and by duration, the darker the fuller; throughput, requests served per 30-second bin. Read the three together: a GAIN in the 99th percentile shows as the candles' upper wicks shortening on the blue side while the heatmap's upper band thins and the throughput bars stand level or higher.

12.12 The governor's ticker

From the repetition's own `audit.jsonl`: the minute and second of the decision, the compass position (0 calm, 1 at the line), and the force on the pedal (push: toward more; lift: handing the setting back toward native). Then the machines it recommends, and the give-back verdict's state: *left native* means every setting stays native's; *acting* means a trial on its own measurement. Every line of the ticker is a line of the governor's audit, and the audit is the record of authority for what it did.

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12.13 Native's own controllers' ticker

Native has no governor; its own controllers' moves are the second ticker, read from the native arm's `capture.csv`: the replicas its pod autoscaler holds, the pods waiting for a place and the machines ready, every change in the same repetition, newest first. With Omni-Compass OFF the first ticker is empty, for nothing decides on top of native. The two tickers side by side are the mechanism of action in its plainest form (chapter 26).

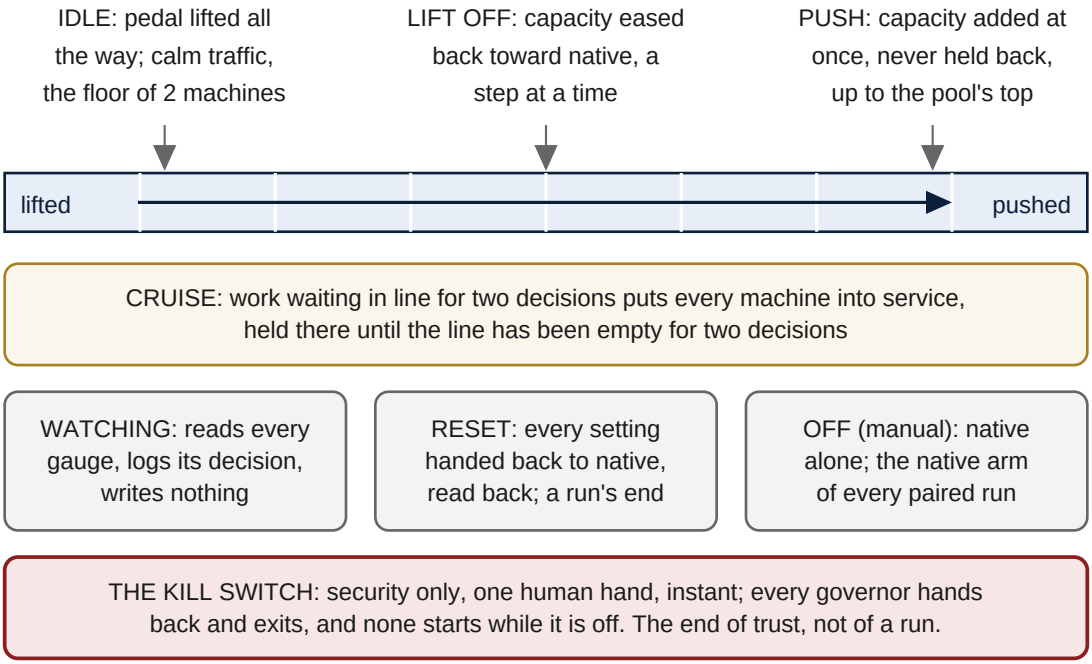


Figure 5. One pedal, no brake. Pushing the pedal adds capacity at once; lifting it eases capacity back toward native a step at a time, never below idle, the floor of two machines. Cruise holds every machine in service while a known queue waits. Watching reads and writes nothing. On the governor's ticker, push and lift are the force column, and the machines recommended are what the pedal bought or handed back.

12.14 Every repetition side by side, and the run's receipt

Each repetition of the run as one small pair of lines; the receipt of this run as the repository wrote it; and this repetition scored by the repository's own scorer, which must equal the run's own summary over its repetitions (the count of matches is printed). Benchmark status, by block: landed, running, queued, or not yet run, for every Kubernetes benchmark of the campaign.

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CHAPTER 13

Kubernetes: Per-Repetition Telemetry

13.3 What it is (the page's section 4d, `#s-runs`)

The benchmark runs themselves at one-second resolution: each three-run table's runs A, B and C read back from the raw files GitHub archived (`results/live/raw`, on the `raw` branch), native and Omni-Compass, repetition by repetition, played second by second. Every file the page reads has its SHA-256 computed and matched against the run's own `SHA256SUMS_ALL.txt`, and every Kubernetes repetition is scored by `tools/live_reps.py`; where a run holds its own scored summary, the scorer's mean over its repetitions equals it. Class L: real Kubernetes on kind, real databases and a real broker.

13.4 How to use it

Choose the benchmark, the run and the repetition; the the page's section 4c dashboard follows. Read a gauge's two lines against each other for the whole repetition, then read the receipt of the run under it: the receipt's word is the reading; the lines are why. A repetition whose job failed and ran again is marked; one that failed three times is listed and left out.

CHAPTER 14

The Shipped Controller's Decision Trace

14.4 The one place the brain drives the muscles decision by decision (the page's section 4e, `#s-live`)

The shipped controller, `omni_controller/controller.py`, on the toy cluster of `pilot/selfpilot.py`, wired as the repository wires it. Each decision runs the chain left to right: the senses, the six-state brain, the nerves and the machine organ's grant, the compass law on the service (the cloud setting), the verdict's trial, the release gate, the plug. Every value below the chain is the controller's own audit

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record of that decision, recorded by `docs/engine/export_controller_trace.py` on runs A, B and C (six hours each, a decision every 60 s, class S). Three buttons choose the setting: *on top of the HPA, the max*; *the cloud: the tune-down, machines only*; and the Linux host and the card (the page's section 4f).

14.5 Reading the trace

The columns, in the audit's order: the stress E, the utility U, the calm, κ, h, σ, ν (the four scalars of the nerves), the gate and whether it passed, whether machines were in service (a blind reading holds the decision), the recommendation, the machines seen, the density ρ, the compass position and velocity, the force, the verdict's word, the writes, the queue, and the law that decided. A row whose gate did not pass wrote nothing; a row whose verdict reads LOSS kept the setting at native; a row with a write names the lever and the value read back.

14.6 The paired change

Per 15-second tick, averaged over ten-minute blocks, with its 95% interval (Student's t) in each run; GAIN or LOSS only when all three runs point the same way with every interval clear of zero. Machines and power: a fall is a cut, a GAIN; a rise is power given. Our own models (class S): a model's small numbers are the model's.

CHAPTER 15

The Chip and the Host

15.4 The highest-value point (the page's section 4f, `#s-stacks`)

The silicon itself, the chip at the far end of every stack: there the max rides each card's own limit, and every watt it takes is power given, shown beside the work it buys. The monitor holds the slots for the metered GPU and CPU runs (class P: the card's own meter on Lambda; the package's own energy counter, `tools/run_cpu_power.py`), every hardware run GitHub has started, newest first, and then the models of one GPU card and one Linux host (class S), native beside Omni-Compass, each run twice on the same load and seed, native first.

15.5 The settings

Buttons choose the stack (a GPU card; Linux, one host; Kubernetes, the page's section 4e; the cloud, the page's section 4a), the load on the card (AI token generation; compute-bound) and the card's power setting (the card as shipped, 150 W; an operator's cap, 105 W). The table under them, *where the mechanism moves on the chip*

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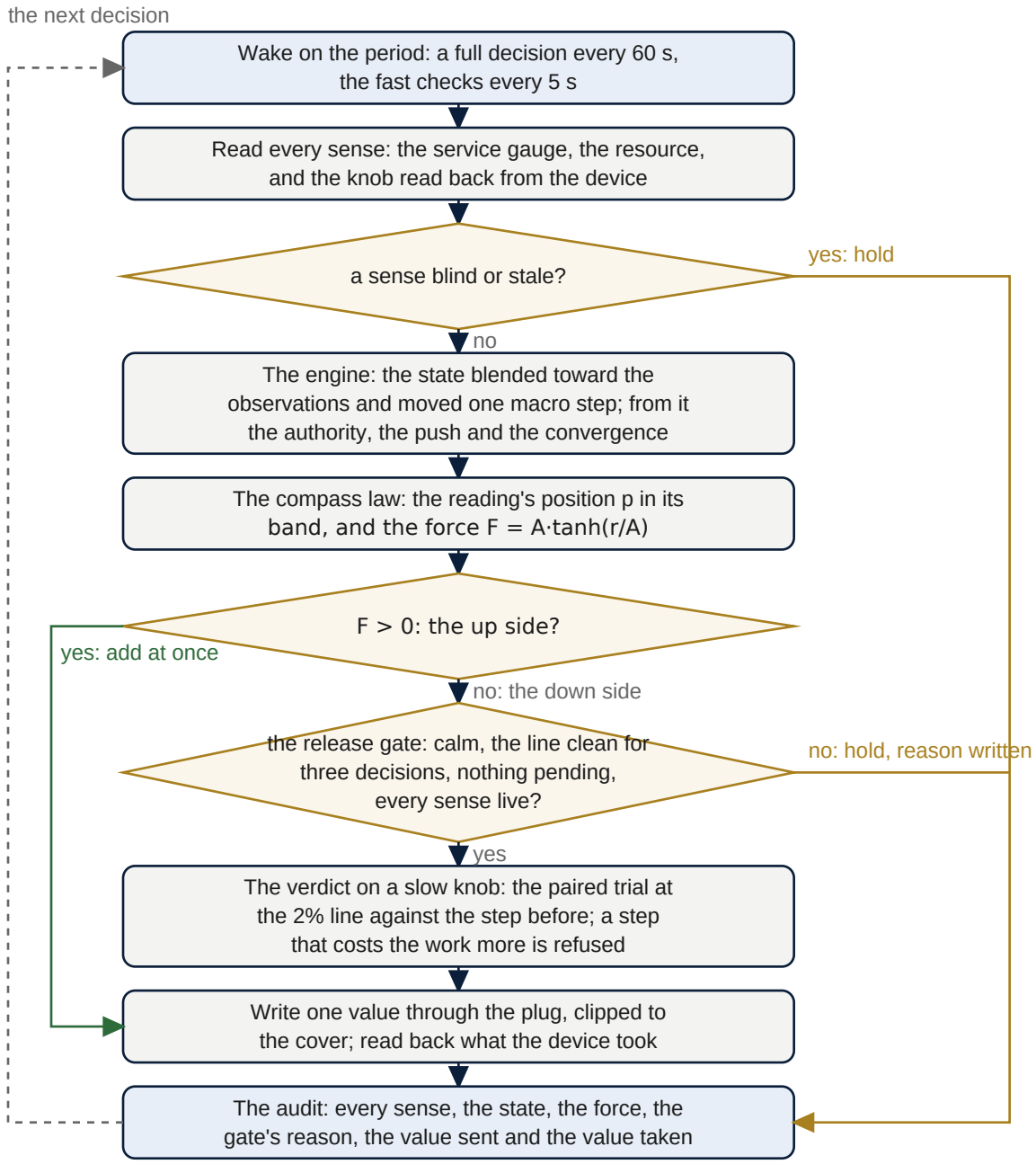


Figure 6. One decision of the governor. Read the body (the senses), place every reading in its band, let the brain's state evolve, grant or refuse through the nerves, apply the compass law to each muscle (push or lift), run the verdict's trial on a step past native, pass the release gate, write through the plug and read it back, renew the lease. On the decision-trace monitor each box is one column of the audit, and *open the same controller, decision by decision* steps through them.

models, lists every setting with three runs each: the seconds held off native, the writes, the power (W) or the host's busy share native · Omni-Compass, and the response (ms) native · Omni-Compass. Click a setting to play it.

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15.6 Reading a row with nothing held off native

A row with no second held off native is Omni-Compass deciding to stay native: its trial of one step read LOSS and was refused, or the knob is already at its limit on native. That row is the engine's own rule at work and is not a fault. The slots marked not yet run are written as the repository holds them and fill in, played back from their raw files like every Kubernetes run, the moment their three runs land.

PART FIVE

The Organism Live and the Console

The organism rendered live, native beside Omni-Compass, its body scan and immune system; the telemetry analytics; and every knob of the console.

CHAPTER 16

The Organism, Live

16.6 The two instances (the page's section 5a, `#s-engines`)

The system rendered as an organism: each node is a muscle (one controlled actuator) and each ring its operating range. The left instance runs under its native controllers alone; the right runs the same controllers with Omni-Compass's brain, nerves, heart and immune system supervising them. Three buttons choose the body: *each muscle alone* (native, then on top, one at a time), *one organism* (the sixty muscles in one body) and the *stress-test model* (the console's sixty muscles). Evidence class S: our own models, shown beside the real runs and never presented as proof.

16.7 Reading a ring

A ring is a muscle's band drawn as a circle: calm at the inner edge, the line at the outer. The muscle's live value is the ball on the ring; the reference, the knob at native, is marked. Under the max a muscle is moved only where it has room above native; a muscle already at its limit on native is left native by design and its ball sits where native's does. The counters under each ring and the scoreboard (chapter 20) give both sides and the difference. Where the two sides look the same, they are the same.

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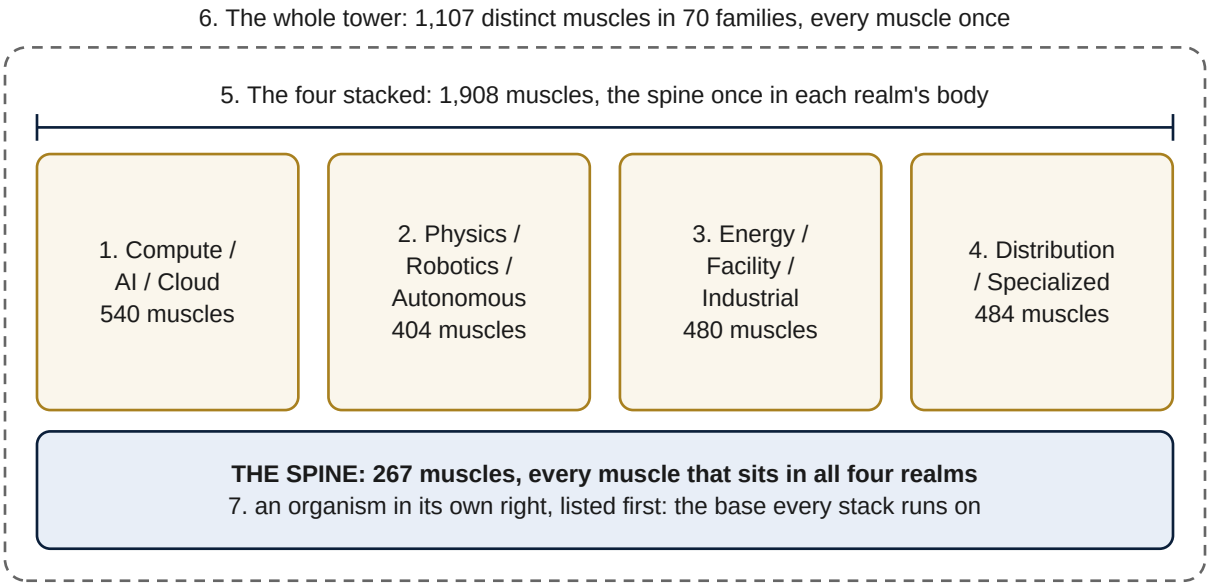


Figure 7. The seven organisms. The spine (267 muscles, every muscle that sits in all four realms), the four realms (Compute/AI/Cloud, Physics/Robotics/Autonomous, Energy/Facility/Industrial, Distribution/Specialized), the four stacked, and the whole tower of 1,107 muscles in 70 families. The organism monitor plays one organism of sixty muscles and each muscle alone; the benefit board (chapter 9) carries all seven.

16.8 The body scan

As a scan reads a living body organ by organ, this reads the system playing in the page's section 4c (the benchmark, run and repetition chosen there), organ by organ, half minute by half minute, from its own SHA-256-checked files: native beside Omni-Compass. Each organ names what it reads; where native has no such organ (the governor's heart, the nerves) the cell says so.

16.9 The immune system, live

The immune system is Omni-Compass's; native runs on its own controllers. Every room and every machine pool has a native twin: the same muscle under its own controller, on native's heat and load, the body's memory of healthy. A room that stands warmer than its twin is inflamed. Where its own knob can cool it, it is cured on the spot. Where it cannot, it sends the whole body a signal, its heat debt, like a cytokine calling for help, and the other muscles answer: the machine pools give back what Omni-Compass added until the room stands no warmer than native's. The skeleton (the bone: the power and cooling lines the whole body shares) grants nothing it cannot carry. A cooling setting is walked back, never snapped. On the monitor an inflamed room is marked, its heat debt is a bar, and the pools' give-back is seen on their rings.

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16.10 The anatomy

Part by part, in the order of control: the brain first, the whole organism last. Each part names what it reads, where it lives in the code, and its measure on this run.

CHAPTER 17

The Telemetry Analytics and Omni-Compass in Its Own Words

17.6 The tensor (the page's section 5b, `#s-lab`)

The engine's own record is a tensor: two sides × every decision × sixty muscles, each entry a muscle's compass position (0 calm, 0.5 the center of its band, 1 its line), recorded from `realms/harness.py` (class S). It fills in decision by decision as the page's section 5a plays. Read it as a sheet of heat: a column (one decision) that is light on the blue side and dark on the amber side is a body Omni-Compass held nearer calm at that moment.

17.7 The difference matrix and the principal motions

The difference between the sides, muscle by decision, with the sign of the word; and the body's principal motions, the tensor's eigen-decomposition: the few patterns of movement that account for most of the body's motion, each with the share of the variance it carries. A first motion that is the whole body rising and falling together is a body breathing with its demand; a motion that is one room against the rest is that room's inflammation.

17.8 The service level and the error budget

The official runs (class L) read as a site-reliability engineer reads a service: the service level this repetition, the error budget burned, and every gauge of this repetition against native.

17.9 Omni-Compass in its own words

Omni-Compass speaks for itself: every line is read from its own records at the moment on screen (the official run playing in the page's section 4c, its governor's own audit, and the engine's harness on the console's clock) and says what it is doing, beside what native does. *Speak aloud* reads it out on the viewer's own browser voice as the run plays.

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17.10 Ask Omni-Compass

Ask in your own words, or pick a question (what are you doing right now; where do you read LOSS, and why; explain yourself to a buyer; show me your largest GAIN). Omni-Compass answers in the first person, for its own whole without it. The answers are drawn from the page's records and are not reviewed by any outside party.

CHAPTER 18

The Control Console

18.5 What the console is (the page's section 6, `#s-desk`)

Runtime parameter control and fault injection on the live engine: load scaling, noise injection, sensor latency, capacity loss, a demand surge, a sensor blackout, a security hold, the dial, and the kill switch. Every control is applied to both sides identically; the response is plotted per decision, native beside Omni-Compass. The engine these knobs turn is the engine's own mathematics running in the page: the brain, the compass law and the verdict, proven equal to the Python and the C++ on every build. It governs the stress-test model's sixty muscles, each on top of its own native controller. The knobs move the live engine in the page's sections 5a and 7a to 7c, and the kill switch and the frame load also move the page's own render governor (the page's section 10).

18.6 Do no harm, at the console

Every control acts on both sides alike; what Omni-Compass does with them is held by its guards: no muscle driven below native or past its own limit, nothing taken from one muscle to give to another on the engine that stands, a give-back tried against native first and refused on a LOSS. Its one meaning is written once, in `DISCLOSURES.md` the page's section 3, and the page's back matter carries it.

18.7 Every control

Appendix B lists each control in a table. In reading order:

Control	Marked	What it does, on both sides
Demand (×)	LIVE	scales every muscle's demand from this decision on; ×1 is the day as written

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Control	Marked	What it does, on both sides
Noise σ	LIVE	multiplies each muscle's demand every decision by 1 + σ·z, z a fresh normal draw, one draw per muscle shared by both sides
Sensor delay (+8 ms, +16 ms, as it is)	LIVE	adds decisions between what a muscle does and what its controller reads, on both sides: native's controllers and Omni-Compass's law read the same late number
Surge ×2	LIVE, 40 decisions	doubles that body part's demand
Lose half its machines	LIVE, 40 decisions	halves a pool's capacity in service
A sense goes blind	LIVE, 40 decisions	forbids any hand-back (blind means hold)
Security hold	LIVE, 40 decisions	forbids any addition
Kill switch	LIVE	every lever to native at once; the window shaded red; the switch at the top of the wall reads red
Turn Omni-Compass OFF	LIVE	the brain reads in shadow; every Omni-Compass side follows native in native's colour
The dial: the max / tune-down	LIVE	the max: every muscle to its own limit, never below native; tune-down: never more than native, capacity handed back only with the organ's grant and the release gate
Pause, one decision, 1×, 4×, 16×, 60×	LIVE	the console's clock: a decision every 0.9 s at 1×; one decision steps once
The same day again, a new day	LIVE	replays the seed with the knobs as set; a new day draws a new seed
Every knob to its start	LIVE	returns demand, noise and delay to the day as written

18.8 What a fault shows

Press *Surge ×2* and watch the page's section 7b: both sides' dots for that body part move up and right; native's controllers each see their own reading one to three decisions late and add in the middle of their range; the

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compass law pushes at the first decision the position leaves the center, so the blue dots turn back before the right edge while amber ones cross it. Press *A sense goes blind* and the governor's hand-backs stop while its additions go on (blind means hold, never a guess). Press *Security hold* and every addition stops while hand-backs go on. In every case the native side is put through the same fault, so the difference is Omni-Compass alone.

PART SIX

The Brain

One control cycle traced, the phase plane and the scoreboard, the brain's state, the verdict, the failure modes, the component specification, and the page that governs its own drawing.

CHAPTER 19

One Control Cycle, Traced

19.3 What it shows (the page's section 7a, `#s-decision`)

One control cycle traced end to end at reduced speed: the sensed inputs, the brain's decision, and the actuator (the muscle) it wrote. The max: every muscle to its own limit, never below native; the capacity it holds is power given. The tune-down: never more than native, and a step back only with the organ's grant and the release gate. In this model native already sits near its band, so the tune-down finds little to hand back; the scoreboard (chapter 20) reads its runs A, B and C live when the dial is on the tune-down. Class S.

19.4 How to read it

Follow the particles: in along the wires from the muscles' meters to the inner arc of nine senses; through the brain's six spokes; out through the nerves (the grant), the compass (the force) and the plug (the write). Each stress on the console lasts 40 decisions from the moment it is pressed, and both sides meet the same demand, so a cycle watched during a surge shows the push on the blue side and the native controllers' late, middle-of-the-range moves on the amber one.

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CHAPTER 20

The Phase Plane and the Scoreboard

20.4 The plane (the page's section 7b, `#s-plane`)

A phase portrait, the picture control engineers use for a system's state. Across is where each muscle sits in its band: p is 0 when calm, 0.5 at the center and 1 at its line. Up and down is how fast it moves there (v, per decision). A muscle at rest sits on the middle line. One climbing toward its line moves up and right, and past the right edge it is past its line. Native is on the left and Omni-Compass on the right: the same muscles at the same instant, one dot each. In the stress-test model the shading on the right is the force the compass law would apply there: the add force toward the line, the hand-back force toward calm. In the realms harness each muscle answers to its own rule, so no single field is drawn.

20.5 The scoreboard

Each panel is one gauge of the live organism, read every decision on both sides, as an operations dashboard shows it: native's value now and Omni-Compass's, the last 120 decisions of both (native solid amber, Omni-Compass blue), and the running paired reading in the tables' words. With the master switch OFF the decisions are shaded red and Omni-Compass's side follows native.

20.6 How each row is read

The paired change, native against Omni-Compass at the same decision, averaged over blocks of ten decisions so neighbouring decisions do not count twice, with its 95% interval (Student's t over the blocks). *This run*: an interval that includes zero reads EVEN; clear of zero, the word follows the gauge's aim (work: a rise is GAIN; misses, failed work, waiting: a fall is GAIN; energy or capacity held: a rise is power given, a fall a cut). *Three runs*: GAIN or LOSS only when all three runs point the same way with every interval clear of zero; EVEN when an interval includes zero; the runs disagree when clear runs point different ways. The spread of positions can widen a little under the tune-down, because the calm muscles it hands back move toward their line while the busy ones stay.

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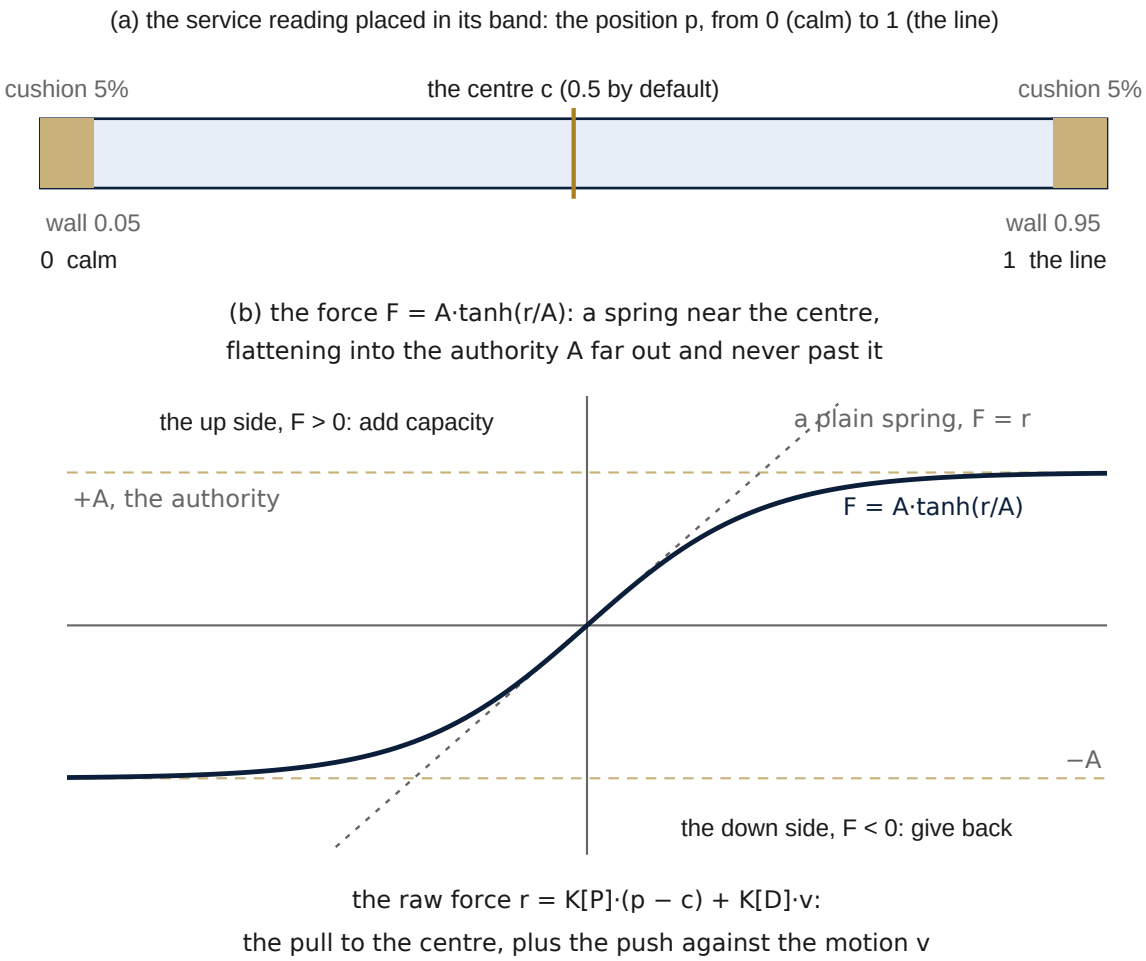


Figure 8. The compass band and the two forces. A reading is placed in its band, p = (r − lo)/(hi − lo), calm at 0, the center at ½, the line at 1, with 5% cushions at each end. Two antagonist forces act on the knob: the add force as the reading climbs toward the line, the hand-back force as it falls toward calm, combined as F = A·tanh((k$_P$(p − ½) + k[d]·v)/A), smooth, never hammering. It fails up past 0.95 and never hands back while it climbs above the center. The phase plane's shading is this force at every point.

CHAPTER 21

The Brain's Internal State

21.5 What is drawn (the page's section 7c, `#s-brain`)

The brain's six state variables rendered on their potential-energy landscapes: a state resting at a minimum is stable; a state driven toward a maximum is under load. Each ball is a live value of the engine's brain,

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integrated by the Runge–Kutta method every decision. In the stress-test model its push and target feed the body's levers. In the harness views the harness arm acts muscle by muscle with no brain, so the brain drawn here is the page's own, fed the harness's readings. Class S.

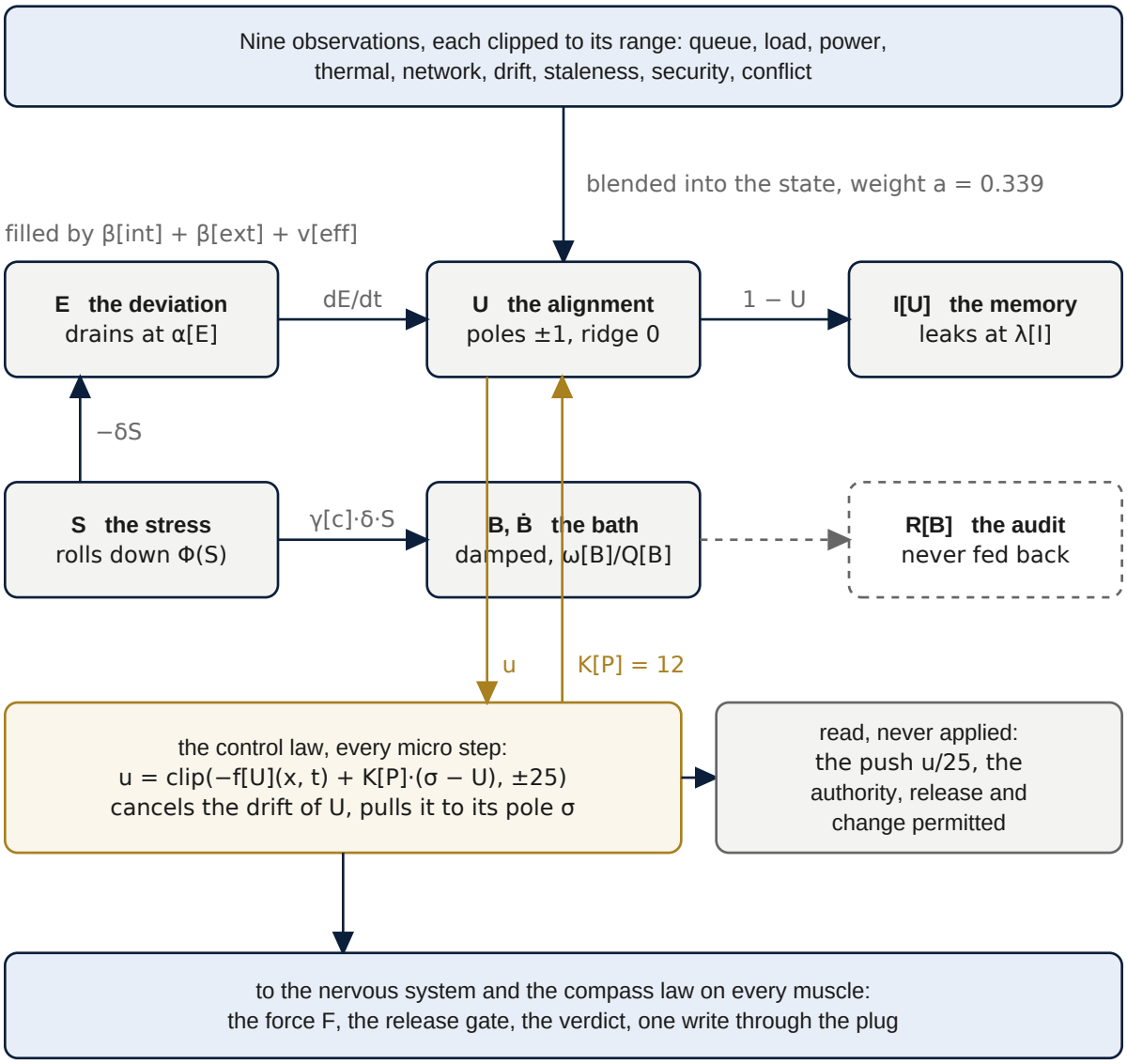


Figure 9. The engine's six states. E the stress, U the utility, I[U] the memory of need, S the stance, B the slow tide of physical load and Ḃ its rate, each drawn on its own scale (E, U and I[U] on [0, 1], S on [−1, 1], B on [−0.5, 1], Ḃ on [−0.5, 0.5]). The brain monitor draws each as a ball on its own landscape, live, every decision.

21.6 The three landscapes the page prints

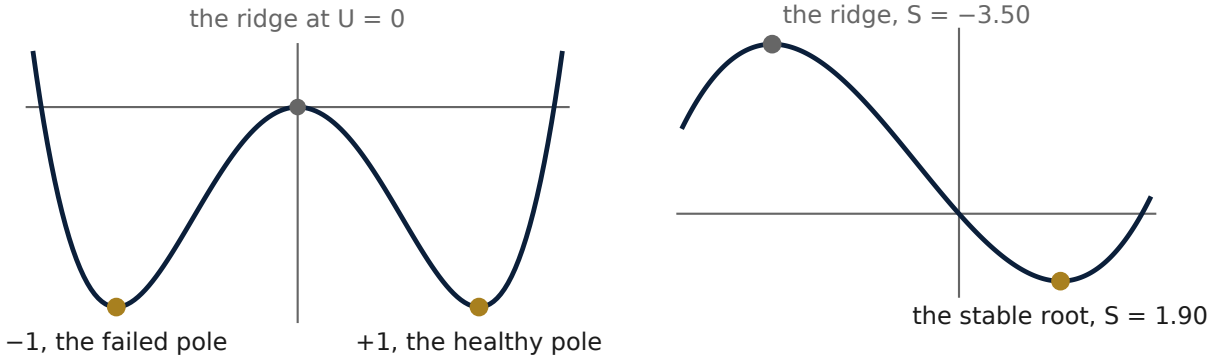
The utility's well, $V(U) = -\mu(U^2/2 - U^4/4) + (\dot{E}/E_{max})\cdot U$: a rising stress E tilts the well toward the basin of collapse, and the ball rolls toward it unless the body is relieved. The stance's well, $\Phi(S) = \alpha_s S^2/2 + \beta_s S^3/4 -$

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δS: S rolls to its minimum S* and the senses pull it back each decision. The tide, $\ddot{B} = \gamma[c]\cdot\delta\cdot S - (\omega[B]/Q[B])\cdot\dot{B} - \omega[B]^2\cdot B$: the slow tide of physical load, a damped oscillator driven by the stance.

(a) the alignment U, a double well:
$V(U) = \mu(U^4/4 - U^2/2)$

(b) the stress S, a cubic potential:
$\Phi(S) = \alpha_s S^2/2 + \beta_s S^3/4 - \delta S$
(α_s 0.12, β_s 0.10, δ 0.5: the engine's defaults)



$dU/dt = \mu U(1 - U^2) + \ldots$
the ball rolls to the nearer pole

$dS/dt = -d\Phi/dS$: S rolls to the stable root;
the landscape is open below the ridge

Figure 10. The double well. The utility U rests in one of two basins: the working basin, where the body does its work inside its lines, and the basin of collapse. The stress E tilts the landscape; the brain's push and the body's relief tilt it back. A ball near the ridge between them is a body under load; the compass law and the nerves exist to keep it on the working side.

21.7 Calm and the grant

calm = $\kappa\cdot h\cdot(1 - (\sigma - 1)^+)\cdot(1 - \min(\nu, 1))$, where κ is the body's coherence, h the headroom, σ the stale share of the senses and ν the conflict. An organ may hand capacity back only when calm ≥ its threshold θ, the line is clean and every sense is live; expanding stays allowed except under a security hold. Under the max, lifting off toward native never needs this grant; going below native does. The monitor draws calm against θ for each organ, so the operator sees why a hand-back was refused.

21.8 Native, beside

Native has no brain to draw: each of its controllers reads one reading, its own muscle's, and moves. The amber panel draws, live from the same decisions, the mean position native's muscles' readings put them at, the share of its muscles past their line, and how many of its own controllers moved their setting this decision. Native has no nerves: nothing grants or refuses a controller, and a blind sense or a security hold changes nothing it does.

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CHAPTER 22

The Verdict

22.4 The change-acceptance test (the page's section 7d, `#s-verdict`)

Before a knob moves one step past native, the step runs as a trial: costs at native first, then at the step. The engine reads the change exactly as a report reads a row, on the log of each cost, with its 95% interval. Each step is kept unless the measurement reads LOSS or its estimated cost exceeds the 2% allowance; an EVEN result (no difference resolvable from the noise) is accepted, so on a high-variance workload a step with a small true cost can pass, and the lab under the monitor measures that false-acceptance rate.

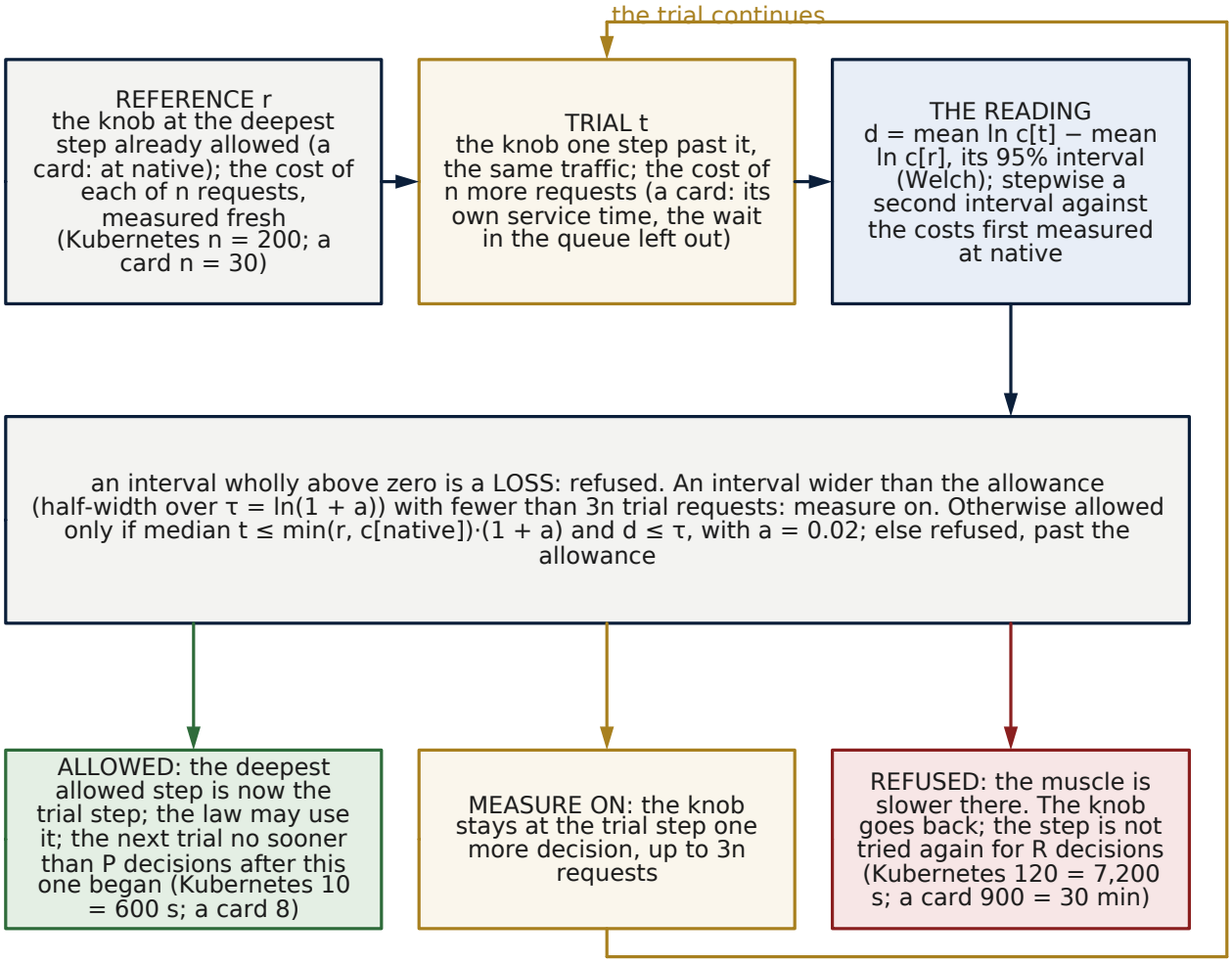
22.5 The lab

Three knobs: *what the step truly does to the cost*, the *allowance* (tolerance) and the *noise σ of the log cost*, with the *pieces of work measured each side*. Set what the step truly costs and watch the rule decide: the keep ratio at each true cost is drawn, beside a rule of medians alone for comparison (the medians alone allow a step when the trial's median is within the allowance of the reference's; the engine's rule also refuses any step whose interval sits wholly above zero, and measures on while the interval is wider than the allowance, up to three times the pieces each side, keeping a step only if both estimates are inside the allowance).

22.6 How each muscle wires it

The Kubernetes controller measures 200 requests each side against a 2% allowance (`--verdict-samples 200 --allow 0.02`), the cost being each request's response time; the noise is measured at every build on each request of the newest tables' own runs. The GPU card measures 30 requests; in its model the card's own time per request carries no noise. The toy gate measures 60 (σ 0.25). Native has no trial: its knob never moves past its own setting, so there is nothing to judge; every step Omni-Compass takes past native has to pass this test first.

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The verdict stands over every slow knob (the machines, the clock ceiling, a cache's notch). A trial begins only while the service is calm and nothing waits; a phase that sees the service leave calm, or that takes longer than its limit (Kubernetes 20 decisions = 1,200 s; a card 30 decisions), is abandoned and the knob goes back, the step neither allowed nor refused. The knob never passes the deepest allowed step, so steps cannot add up past the allowance. On the max, the Kubernetes machine count never gives back a machine the cluster had on its own.

Figure 11. Block diagram: the verdict's paired trial around a slow knob, read GAIN, EVEN or LOSS. The reference is the cost of n pieces of work at the deepest step already allowed; the trial is the cost of n pieces at the step. Both on the log of the cost, with the 95% interval of their difference: an interval wholly above zero refuses the step; one inside the allowance keeps it; one wider than the allowance measures on, up to three times n each side. The verdict monitor runs this rule live at any true cost the viewer sets.

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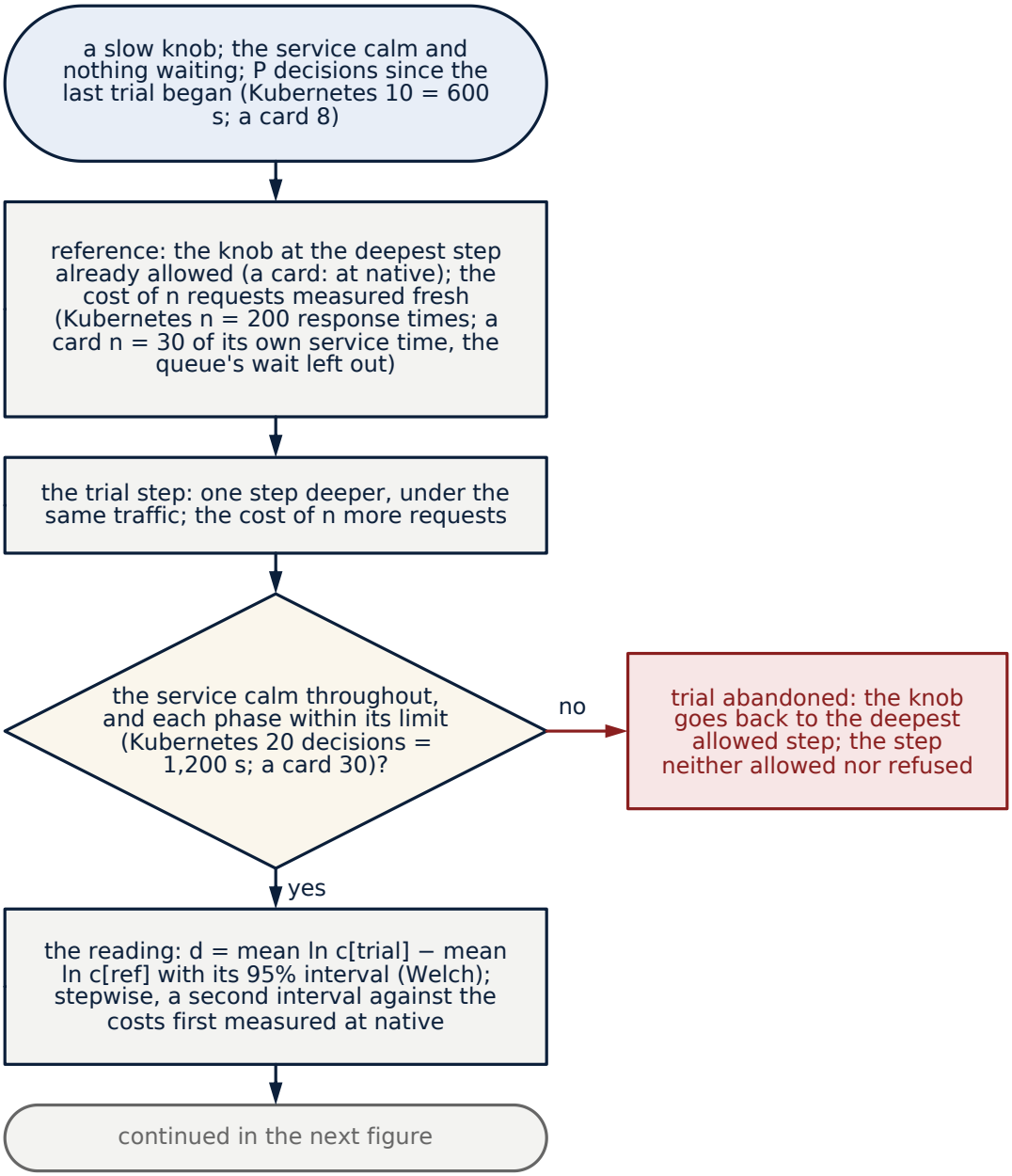


Figure 12. Flowchart: the give-back trial. A trial starts only with the organ's grant and a clean line; it runs to its full measurement and is never ended by the calm it causes; its reading against native decides, GAIN or EVEN keeps the step, LOSS refuses it and the setting stays; the release gate then passes or holds the write. The ticker's state column reads left native until a trial starts and acting while one runs.

CHAPTER 23

Failure Modes, Controls and Residual Risk

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23.3 The register (the page's section 8, `#s-fail`)

Every known way Omni-Compass can fall short, the control that mitigates it, that control's coverage and the residual risk where it does not reach, and its implementation in the code; every failure mode the live engine can be put through has a fault-injection control on the console, named in the row. The register is `docs/SAFETY_CASE.md` and `docs/THREAT_MODEL.md` read as a table.

23.4 The resource costs shown, not judged

Under the register, the rows the three-run tables report but do not judge: the governor's own commands, its processor time, the pending pods in a snapshot, the energy per core-hour. They are printed so the cost of the governor is never hidden; they carry no word because their preregistrations declared them shown only.

CHAPTER 24

The Component Specification

24.3 Every part and its equation (the page's section 9, `#s-math`)

The component specification: every part of the engine, its equation in Unicode as the repository writes it, and the source module that implements it. Items 1 to 5 and 8 are the engine's code and run in every view; items 6 and 7 as written (the compass law on each muscle, the plug's lever) drive the stress-test model; in the harness views the max and the plants' own overrides take their place; the live controller runs the compass law on the service, as the page's section 4e shows.

Item	Part	What the page prints	Module
1	Senses	the muscles' meters read and normalised to nine observations: q, load, power, thermal, network, drift, stale, security, conflict	`adapter.observe_vector`
2	Assimilate	$x \leftarrow (1 - a)\cdot x + a\cdot x[obs]$, $a = 0.339$: the engine is pulled toward what the senses read	`adapter.assimilate`
3	Evolve	equations (1) to (7), the Runge–Kutta method, ten micro steps of 0.01, the controller evaluated and not applied	`core.macro_step`
4	Allocate	$push = clip(-f[U] + 12\cdot(1 - U), \pm 25)/25$ and $\rho^* = clamp(\rho_0 - k[I]\cdot I[U] - k[E]\cdot E, \rho_{min}, \rho_0)$	`adapter.Governor.step`

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Item	Part	What the page prints	Module
5	Nerves	κ, h, σ, ν into calm; calm against each organ's θ grants contraction (the spine to nodes, θ 0.7; compute to pods, 0.5; physics to the processor's clock, 0.3; energy to power, 0.3; distribution to routing, 0.5)	`nervous_system`
6	Compass	p = (r − lo)/(hi − lo); F = A·tanh((kₚ(p − ½) + k[d]·v)/A); fails up past 0.95; never hands back while climbing above the center; k[d] at three times critical damping from the muscle's own τ, the push	`compass_law.CompassLaw`
7	Plug	one wire in, one wire out: each write clipped to the cover; the snapshot taken once at the start is what reset restores; each write read back; the kill switch hands every lever to native at once	`compass_law.Plug`, `master.py`
8	Verdict	the paired trial of chapter 22	`verdict.py`

24.4 What the specification is for

It lets the reader of the page check that what moves on the monitors is what the code says: the six spokes at the center of the cycle (E, U, I[U], S, B, Ḃ) are item 3's state, the inner arc of nine segments is item 1's senses, the force field of the phase plane is item 6's law. The Owner's Manual carries the derivations; this page carries the statements.

CHAPTER 25

The Page Governs Its Own Drawing

25.3 Omni-Compass inside the page (the page's section 10, `#s-self`)

The engine that runs the page has Omni-Compass in it. Native is the browser's own loop: every chart on screen drawn on every frame, and when the frame overruns, whole frames dropped. Omni-Compass is the same loop with the engine's own compass law governing it. Each chart is a muscle whose knob is how many frames it waits between draws, and the reading is the time between frames. Below the band's center the pedal lifts and every chart is handed back to native, drawn every frame. Toward the line it pushes: the charts farthest from

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where you are looking wait a frame or more, so the frame itself stays inside its line. Evidence class L: live software, on the viewer's own machine.

25.4 Reading it

The monitor shows the frame time on both sides, the frames dropped by native's loop, and the frames waited by each chart under Omni-Compass. On a laptop with the wall in view, the charts on screen draw every frame and the rest wait; scroll, and the waiting moves with the eye. The kill switch on the console reaches this governor too: engage it and the page draws as the browser alone would.

PART SEVEN

Reading a Result off the Page

The mechanism of action side by side, the trace from a tile to its record, what to do when a tile reads LOSS, and what the page does not tell you.

CHAPTER 26

The Mechanism of Action, Side by Side

26.5 What native does

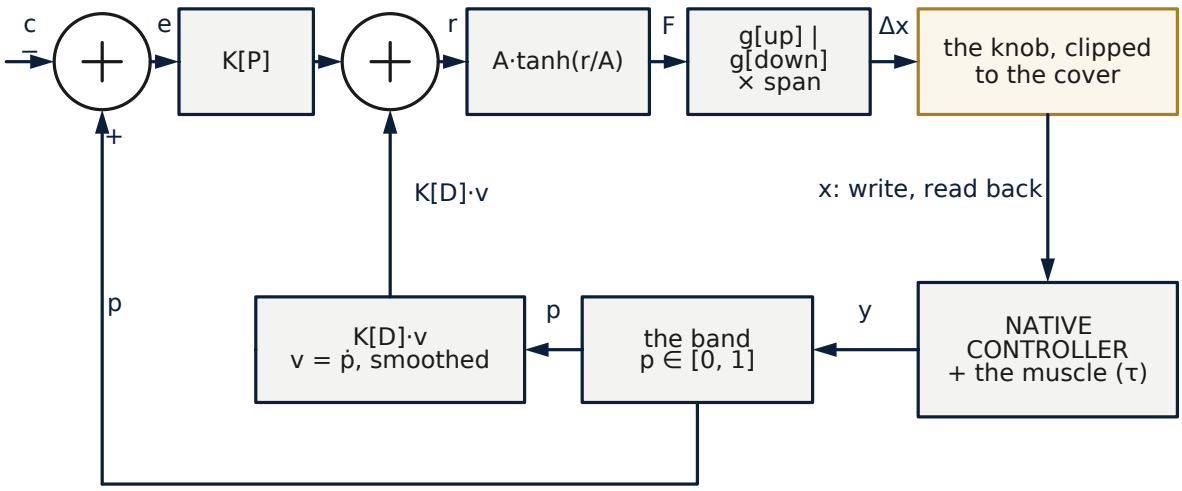
Each native controller reads one reading, its own muscle's, one to three decisions late (the pod autoscaler one sync every 15 s), and moves its one setting toward the middle of its range, because protecting its one thing is its whole job. It has no heart (nothing proves it is still there), no brain (no state, no memory of need, no stress, no tide), and no nerves (nothing grants or refuses it, and a blind sense or a security hold changes nothing it does). Its moves are the second ticker of the page's section 4c and the amber panel of the page's section 7c.

26.6 What Omni-Compass does, on top

Once every decision it reads the whole body through the plug (the senses), places every reading in its band, lets the brain's state evolve, grants or refuses through the nerves, applies the compass law to each muscle (push toward more as the reading climbs, lift toward native as it falls, smooth, never hammering), runs the verdict's trial on any step past native, passes the release gate, writes through the plug and reads it back, and renews its lease. Every one of those is a column of the audit, a line of the first ticker and a box of the cycle monitor. The

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native controllers keep running underneath, untouched; Omni-Compass moves only the settings they already expose.



c the centre of the band; y the reading; p the reading's position in the band; e = p − c; v = ṗ smoothed; r the raw force; F the force after the saturation; Δx the step; x the knob. Fail up: p ≥ 0.95 → F = A at once; the down side held until p is back inside the band. No take-back while the reading climbs above the centre (F < 0 with v > 0 and p > c → F = 0). K[D] is set from the muscle's own response time τ (critical damping), never tuned by hand.

Figure 13. Block diagram: the compass control loop. The muscle's reading comes in on the sense terminal, is placed in its band, and the compass law computes a force; the force is a setting written out on the motor terminal, inside the knob's cover, through the plug, and read back; the native controller sits inside the loop, still reading its own gauge and still moving its own setting; the kill switch opens the motor terminal and the lease restores the snapshot. On the page the two tickers show the two loops, native's own and the governor's on top of it, at the same seconds.

26.7 Where to see the difference

On the official run playback (chapter 12): the first ticker shows the pedal pushed at the compass position's first move off the center and the machines recommended before native's first late sample; the second ticker shows native's autoscaler adding replicas a sync or two later. On the phase plane (chapter 20): the blue dots turn back before the right edge where the amber ones cross it. On the scoreboard: the work waiting lower, the decisions past their line fewer, the capacity held higher during the climb (power given) and eased back after it (the lift). On the brain (chapter 21): calm falling under a surge and the grant refused until the line is clean again, which native has no equivalent of.

26.8 What the difference is not

It is not a competitor's dashboard. Omni-Compass never replaces the controller it sits on; every move native makes is still made. A monitor where the two sides are identical is a muscle Omni-Compass left native, at its limit or with nothing to add, and that reading is EVEN and correct.

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CHAPTER 27

From a Tile to Its Record

27.3 The chain

Every number on the wall can be walked back to the file it came from.

1. **The tile** names its gauge and its word, and links to the monitor it copies.
2. **The monitor** names the benchmark, the run (A, B or C) and the repetition, or the table and the row, and the evidence class.
3. **The three-run table** is in `results/live/` (`V13_*.md` on the engine that stands; each table names its engine), with its runs' ids and its rule printed at its head.
4. **The run's raw files** are on the `raw` branch at `results/live/raw/run-<id>/`, with `SHA256SUMS_ALL.txt`; the page matched every file it drew against that list. `scripts/raw_archive.sh open /run-<id>/` places them where the table names them.
5. **The governor's audit** is the run's `audit.jsonl`; native's own controllers' moves are its `capture.csv`.
6. **The scorer** is `tools/live_reps.py`; the page's per-repetition score equals the run's own summary over its repetitions.

27.4 The engine's name

Every table and every monitor names the engine it ran on; the page's title block names the engine the page stands on (omni-v13). A table on an older engine stands only until its runs on the engine that stands land, and the board says which it is. Nothing is read across engines.

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CHAPTER 28

When a Tile Reads LOSS

28.6 The rule

Theoretically the word LOSS should never appear: the engine that stands moves a knob only where a paired trial on the muscle itself shows the muscle never behind for it, and where it cannot bring a knob a gain, that knob stays native. When LOSS does appear, the row stays, and the founder's order governs: if Omni-Compass hurts something, no mechanism of Omni-Compass is pushed through that muscle. The cause is found in this order, our own wiring first, native's setup second, the scoring third; the mechanism or its wiring is corrected; the test runs again on the next engine version, and the earlier table stands as its engine's record.

28.7 Read the three-run reading, not the one run

A red tile from one run of a table still running (the Azure steady set at 33 of up to 100 repetitions) is that run's own word for that run and not yet a reading: its interval may include zero (EVEN so far), and a three-run word needs all three runs. The alarms table grades it WARN. Wait for the look.

28.8 Open the root-cause document

Where the repository has found the cause it is linked from the tile: `docs/POD_START_WAIT.md` (a pod's start on one shared host is the test machine's own unpacking of an image, the same on both sides at the same host load), `docs/TAIL_LATENCY.md` (where a 95th or 99th percentile reads higher, and why), `docs/BATCH_QUEUE_PREREGISTRATION.md` amendment 11 (the conveyance on a shared host beside another workload's room, fixed in omni-v13). A tile with a root-cause document is a known cause with its fix landed or on the next version's list.

28.9 Which knob, and what to turn

The LOSS row names its gauge; the governor's audit names the knob that moved in that window and why. Three outcomes follow, and only these:

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What the audit shows	What it means	What is done
a knob moved and the gauge went against its aim while the muscle stayed inside its limit	the wire or the rule is wrong for this muscle	the mechanism is corrected in the toy gate first, lands as the next engine version with no LOSS, and the benchmark runs again
a knob moved and the muscle went past its own limit, or another muscle's room was taken	a guard did not hold	the guard is corrected (as omni-v13 did for the conveyance on a shared host) and lands the same way
no knob moved, and the gauge still differs	the test machine, native's setup or the scoring	the rig or the scorer is fixed; a test machine's limit is never reported as Omni-Compass's result

Nothing is ever tuned to make a row read well; no row is steered. What is held to the letter is the mechanism: the code does what its rule says, the wire is fastened right, the harness measures as a deployment runs. The per-muscle native switch, by which an operator leaves one muscle native while the body is governed, is the next version's work (`docs/CHANGE_CONTROL.md`).

28.10 On your own system

The Operator's Manual's "wired right or wired wrong" (the Operator's Manual's section 4.5) and its drill book (the Operator's Manual's chapter 28) are the procedure; `DISCLOSURES.md` the page's section 4 is the rule: where a result elsewhere differs from the published record, the first presumption is the wiring, the settings or the reading, not the law.

CHAPTER 29

What the Page Does Not Tell You

29.7 It decides nothing

The page is a monitor. Every word on it was decided in the repository when its table was written. A reading you disagree with is taken up with the table and its runs, never with the page.

29.8 Class S is not proof

The harness, the shipped controller on the toy cluster, the GPU card, the Linux host, the realms table, the grid's modelled half and the orchestra are our own models. They show the mechanism; they do not measure a

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machine. A model's energy is not a meter's. The page says the class on every view, and the manuals say it again.

29.9 Slots not yet run are empty on purpose

The metered chip runs, the CPU host's meter, the clouds' sets still running and the register's third-party slots are written as the repository holds them and marked not yet run. An empty slot is a true reading of the record. Nothing is drawn in its place.

29.10 The page's age

A page is true for the build it names. The last tile of the wall is its age; the title block is its commit. A reading landed after the build is on the next page.

29.11 The page's Chapter 10 is measured on your machine

The self-governed rendering is class L on the viewer's own device; its numbers are the viewer's browser's and prove nothing about any server.

29.12 Ask Omni-Compass is composed

Its answers are drawn from the page's records by a language model on the viewer's own account, are not reviewed by any outside party, and are not a record. The records are the tables and the raw files.

PART EIGHT

License and Your Copy

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CHAPTER 30

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CHAPTER 31

Verifying Your Copy of the Page

31.5 The engine

`python3 tools/omni_version.py` prints the engine the checkout stands on and its digest; the page's `<html data-engine>` must name the same. `python3 docs/engine/build_live_engine.py --check` rebuilds the page in memory and exits 1 if any record differs from the page; nothing is written.

31.6 The mathematics

`python3 docs/engine/check_live_engine.py` runs the page's brain, compass law and verdict against the Python and the C++ (`docs/engine/cpp_core_probe.cpp`) and prints the largest difference, which is at machine precision.

31.7 The records

`python3 docs/engine/check_records.py` reproduces every record the page carries by an independent path and compares value by value; `docs/engine/APPRAISAL.md` is the standing result.

31.8 The whole

`python3 verify.py` runs every check of the repository, this page's included, and ends `VERIFICATION: PASS`; `python3 tools/release_manifest.py` writes `RELEASE_MANIFEST.json` at the commit and must read 0 FAIL. A copy whose page, engine or records differ from what these print is not a copy of the record.

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Back Matter

BACK MATTER

Glossary

- **Alarm tile.** A judged row of a real or modelled run that read LOSS, or a run that ended without a measurement, raised on the wall worst first with its source and its root-cause document.
- **Anchor.** The part of a page address after # (`#s-health`) that names a section and does not move when sections are renumbered.
- **Band.** A reading's safe range, from calm (p = 0) through its center (p = 0.5) to its line (p = 1), with 5% cushions.
- **Beat.** One turn of the governor's loop: read, decide, write, renew the lease. The heart.
- **Calm.** $\kappa \cdot h \cdot (1 - (\sigma - 1)^{+}) \cdot (1 - \min(\nu, 1))$: the nerves' one number, against which each organ's threshold θ grants a hand-back.
- **Class (P, L, S).** The evidence class of a number: a physical meter, live software, our own models.
- **Compass position (p).** Where a reading sits in its band; 0 calm, 1 at the line.
- **Console.** The page's Chapter 6: every runtime control of the live engine, acting on both sides alike.
- **Decision.** One turn of the governor; the unit of time on every live monitor.
- **Dial.** The max (every muscle to its own limit, never below native; the default) or the tune-down (never more than native).
- **Engine that stands.** The engine version on main, named on the page's title block and in `python3 tools/omni_version.py`.
- **Golden signals.** Latency, traffic, errors and saturation: on the wall, the work waiting, the work done, the decisions past their line and the capacity held.
- **Heart.** The governor's beat and its lease, drawn on the official run playback.
- **Immune system.** Omni-Compass's: native twins, the heat debt, the cure on the spot, the pools' give-back, the bone's grant.
- **Kill switch.** The instant hand-back of every lever to native, for security; red on the switch and on every time axis.
- **Lease.** The governor's proof of life; when it ages past its limit the watchdog walks every setting back to native.

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- **Live / Recorded.** A knob marked Live acts on the live engine; one marked Recorded data chooses a record and modifies nothing.
- **Master switch.** The one switch at the top of the organism: ON (green), the kill switch engaged (red), OFF (white).
- **Monitor.** One section of the page that draws a record or the live engine; its small copy on the wall is a tile.
- **Native.** The whole system by itself with everything it ships with; amber.
- **Omni-Compass.** Omni-Compass on top of native; blue.
- **Pedal.** Push for more, lift for less; the force column of the governor's ticker.
- **Power given.** A resource spent or held that rose: what Omni-Compass put in to buy the work and speed beside it; green.
- **Shown, not judged.** A gauge declared in advance as shown only; grey; counted in no total.
- **Stopping rule.** The cloud sets' preregistered looks at 10, 20, 40 and 70 repetitions with 99% intervals (`tools/stop_rule.py`).
- **Ticker.** The governor's decisions (first ticker) and native's own controllers' moves (second ticker), newest first.
- **Tile.** One small monitor on the wall; click it to go to the monitor it copies.
- **Verdict.** The paired trial every step past native must pass: kept unless it reads LOSS or exceeds the 2% allowance.
- **Wall.** The status board: every live monitor in one view, the switch and the engine first.

APPENDIX

Appendix A. The Page, Section by Section

Page section	Anchor	Title	Chapter of this book
i	`#notice`	Legal notice	Notice and Disclaimer
ii	`#contents`	Engine table of contents	2
iii	`#title-page`	Title page and build	1, 27
iv	`#s-read`	Executive summary and reading key	the key; 3

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Page section	Anchor	Title	Chapter of this book
1	`#p-health`	The monitor wall: system health, now	4
1a	`#s-health`	Status board and the monitor wall; the master switch; alarms; the feed; the event log; the exposition	4, 5, 6
1b	`#s-github`	CI pipeline: GitHub Actions	7
2	`#s-whole`	The whole organism: the synopsis of everything	8
3	`#s-fleet`	Measured effect: every domain; the grid; every three-run table; every muscle and the seven organisms; every family	9
4	`#p-evidence`	Evidence by stack, from the cloud down to the chip	10 to 15
4a	`#s-cloud`	Cloud providers: Azure, Google Cloud, Amazon Web Services	10
4b	`#s-orch`	Orchestrators and governors	11
4c	`#s-ops`	Kubernetes: official run playback	12
4d	`#s-runs`	Kubernetes: per-repetition run telemetry	13
4e	`#s-live`	Kubernetes and cloud: the shipped controller's decision trace	14
4f	`#s-stacks`	Hardware: the GPU card and the CPU host	15
5	`#p-organism`	The organism, live	16, 17
5a	`#s-engines`	The organism, live: native and Omni-Compass; the body scan; the immune system; the anatomy	16
5b	`#s-lab`	Telemetry analytics and Omni-Compass in its own words	17
6	`#s-desk`	Control console	18
7	`#p-brain`	Control architecture: the brain	19 to 22

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Page section	Anchor	Title	Chapter of this book
7a	`#s-decision`	One control cycle, traced	19
7b	`#s-plane`	Phase plane and scoreboard	20
7c	`#s-brain`	The brain's internal state	21
7d	`#s-verdict`	The verdict: change acceptance testing	22
8	`#s-fail`	Failure modes, controls and residual risk	23
9	`#s-math`	Component specification and mathematics	24
10	`#s-self`	Self-governed rendering	25
v	`#s-disclose`	Disclosures, governing law and contact	29, 30
vi	`#notice-end`	Closing legal notice	Notice and Disclaimer

APPENDIX

Appendix B. Every Control of the Console

Control	Where	Acts on	Leaves alone
Kill switch	the top of the wall; the console	every lever of every governor on the page, to native at once; shades every time axis red	the records: nothing recorded is modified
Turn Omni-Compass OFF / ON	the console	the brain's outputs: OFF, read in shadow, nothing leaves; ON, outputs reach the muscles	the native side, which never stops

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Control	Where	Acts on	Leaves alone
The max / Tune-down	the console	the dial of the live engine's governor	the records; the official runs' own dial, named in each
Demand ×	the console	every muscle's demand, both sides, from this decision on	the seed
Noise σ	the console	each muscle's demand each decision, 1 + σ·z, one draw shared by both sides	the seed's day
Sensor delay (as it is, +8 ms, +16 ms)	the console	the decisions between a muscle's act and its controller's reading, both sides	the demand
Surge ×2 / Lose half its machines / A sense goes blind / Security hold	the console	one body part for 40 decisions, both sides	every other part
Pause / One decision / 1× 4× 16× 60×	the console	the console's clock	nothing else
The same day again / A new day	the console	the seed replayed or redrawn, with the knobs as set	the knobs
Every knob to its start	the console	demand, noise and delay, back to the day as written	the dial and the switch
A / B / C	the page's sections 4c, 4d, 4e, 4f	which run is played	the other runs
Benchmark, run, repetition	the page's sections 4c, 4d	which record is played	the record itself
Each muscle alone / One organism / Stress-test model	the page's section 5a	which body the organism monitor plays	the console's knobs, which act on whichever plays
Its own load / The climb / Work waiting in line	the page's section 3	which load the realms table is read under	the table
On-site bare metal / Through the cloud	the page's section 2	which way the stack is bought in the composition	the receipts

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Control	Where	Acts on	Leaves alone
Every measured piece / The strict reading	the page's section 2	which rule the synopsis counts by	the factors
Its one run beside the count / in the count	the page's section 2	whether a governor that replaces a controller enters the whole	the governor's own table
ALERT / WARN / INFO; All / LOSS / The runs disagree / Live now / Guards and open items / GitHub	the page's sections 1a, 3	which rows are shown	the rows
Copy the exposition	the page's section 1a	the clipboard	the page
Speak aloud / Ask Omni-Compass / Stop	the page's section 5b	the viewer's browser voice; a question composed on the viewer's own account	the records

APPENDIX

Appendix C. The Gauges on the Wall

Gauge	Side	Aim	GAIN reads	Power given reads	Where it comes from
The work waiting	both	time waited: lower	the blue line under the amber	—	the live engine; on the official runs, pending pods and the queue
The work done	both	work done: higher	the blue line over the amber	—	requests served, jobs finished, work inside the line
The decisions past their line	both	missed: lower	fewer blue past the line	—	the compass positions over 1
The capacity held	both	a resource held	a fall (a cut)	a rise	machines, replicas, clock in service

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Gauge	Side	Aim	GAIN reads	Power given reads	Where it comes from
Response time, mean, 95th, 99th	both	time waited: lower	a fall	—	the runs' own samples, binned
Time over the response line	both	missed: lower	a fall	—	the share of samples over the line
Failed requests	both	failed: lower	a fall	—	the load generator's record
Alert minutes	both	time over the line: lower	a fall	—	whole minutes whose 95th percentile, failed requests counted over, was over the line
Machines billed, the compute bill	both	a resource spent	a fall	a rise	the cloud's own count of machines, priced at list
Energy (declared model, the meter, Kepler)	both	a resource spent	a fall	a rise	the declared model; the RAPL counters where a meter exists (P); Kepler shown, not judged
HPA replicas, pods started	both	a resource held	a fall	a rise	native's autoscaler's own record in both arms
Pod start wait, total and mean	both	time waited: lower	a fall	—	the pods' own start times; the test machine's limit never read as the governor's
The governor's commands, its additions inside the line, its own processor time	Omni-Compass	shown, not judged	—	—	the audit; native is 0
Frame time, frames dropped, frames waited	both	time waited: lower	a fall	—	the viewer's browser (L, the viewer's machine)

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APPENDIX

Appendix D. The Colours and the Marks

Mark	Meaning
Amber	native
Blue	Omni-Compass
Green	GAIN, and power given
Yellow	EVEN
Red	LOSS; the kill switch and its window; the master switch engaged
White, dashed edge	the runs disagree; the runs differ; not a reading
Grey	shown, not judged; a monitor with no record yet
Dashed blue on amber	Omni-Compass's side following native with the switch OFF
Accent border on a tile	the monitor it copies is on screen now
Red shading across a time axis	a kill-switch window
A bar on a ring	a room's heat debt (the immune system)

APPENDIX

Appendix E. Troubleshooting the Page

Symptom	Cause	What to do
The page is slow or stutters	the browser's own loop would draw every chart on every frame	the page's section 10 governs it: charts off screen wait; close other tabs; the kill switch on the console shows native's loop for comparison
A tile is grey and says "not yet run"	the record does not exist yet	nothing: an empty slot is a true reading; the board says where it stands
Symptom	**Cause**	**What to do**
A monitor says "stopped before a result"	a cloud or chip run ended without a measurement	the alarms table carries it; the run's own log says why (a key cut short, a

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Symptom	Cause	What to do
		plan, a quota); fix at the cloud, never on the page
The age of the page is over three hours	the rebuild did not run or did not land	`.github/workflows/live-engine.yml` on GitHub; `python3 docs/engine/build_live_engine.py` locally
A number on the page differs from a table	the page was built before the table changed	the age tile; the table governs; the next build carries it
The two sides are identical on a monitor	the muscle is at its limit on native, or Omni-Compass left it native	EVEN, correct; chapter 26.4
The first ticker is empty	the switch is OFF or the kill switch engaged	nothing decides on top of native while it is; turn it ON
The page refuses to build	the checkout's engine is not the one the page names	a new engine is written into the page's words by a person first; `python3 tools/omni_version.py`

APPENDIX

Appendix F. The License

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 ↪ rms, riders and statements of work.
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 ↪ Omni-Compass compass rose with its ring of Greek
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APPENDIX

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APPENDIX

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APPENDIX

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4. **Confidential filings.** The patent application number and other confidential information are disclosed to a licensee under the agreement's confidentiality terms.

I.18 Limits of liability

Each party's liability is limited to the fees paid or payable in the twelve months before the claim, and neither party is liable for indirect, incidental, special, consequential or punitive damages, except for the exclusions the signed agreement states (for example, breach of confidentiality, the infringement indemnity, and the licensee's breach of the license scope).

I.19 Confidentiality, data and security

1. **Confidentiality.** Mutual, for the term and a period after it, covering non-public information either party gives the other.
2. **Data.** Omni-Compass contains no telemetry. The licensee's data, readings, settings and receipts stay on the licensee's systems and remain the licensee's. Where the licensee chooses to share data for support, a data-processing agreement applies.
3. **Security.** A software bill of materials ships with every version (`sbom/`); vulnerabilities are handled under `SECURITY.md`; the licensee may run the verifier (`python3 verify.py`) and the seal checks on every delivered version.

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I.20 Safety and operation

1. The licensee remains the operator of its systems. The machines' own controls and protections stay in place; Omni-Compass sets only values they already accept.
2. Any connection to a physical system (grid, plant, vehicle, robot, medical device, facility) requires a written safety case accepted by the system's owner and every review its regulator requires; safety-critical systems are licensed only on those terms.
3. The kill switch stays operable at all times.

I.21 Compliance, audit and insurance

1. Each party complies with export-control, sanctions and anti-corruption law.
2. The licensee reports usage against the licensed scope as the order form provides, and The Omni-Compass LLC may verify it on reasonable notice, no more than once a year.
3. Insurance, where the order form requires it, is carried by each party at the limits stated there.

I.22 Term, renewal and transition

1. Term and renewal are as the order form states.
2. On termination or expiry, the licensee stops use, every setting Omni-Compass changed is handed back to native (the hand-back is built into the engine), and every copy is returned or destroyed; transition assistance is available at the rates in the order form.
3. Source-code escrow for the licensed version is available under a three-party escrow agreement.

I.23 Governing law and precedence

The signed agreement states its governing law and venue; the standard is the law of the State of Delaware, with venue in the Delaware state courts and the United States District Court for the District of Delaware. Its order of precedence is: the order form, then any rider, then the Enterprise License terms, then the documentation. This page and the evaluation license (`LICENSE`) are superseded by the signed agreement for the licensee and its scope.

I.24 How to obtain one

The Omni-Compass LLC, main office infrastructure, toll-free direct 1-888-339-9551, system route asset 1-888-DEX-YLL-1, landline 321-746-4411 registered in Florida, www.omni-compass.com. A license starts with an evaluation on the licensee's own systems (`LICENSE`, section 1.4): watch mode, the wire check, and paired receipts on its own load.

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APPENDIX

Appendix J. Patents and Trademarks

J.3 Patents

All patents, copyrights and trademarks covering the Omni-Compass engine, its mathematics and its software have been filed in the United States by The Omni-Compass LLC.

The filings cover, among other things, the Omni-Compass engine (the six-state closed law and its bounded control command held through the integration step), the compass law (the service position pulled to the middle of its band by a bounded push and pull, with fail-up at the wall), the antagonist up/down wiring of each muscle, the plug contract (one restore point, read-back, the one-writer rule), the two-wire governance of a processor (clock ceiling and power limit), the two-way nervous system that grants and withdraws authority per organ, and one governor over many machines and realms on one clock.

J.3.1 What has been filed

Application	a nonprovisional utility patent application under 35 U.S.C. 111(a), filed with the United States Patent and Trademark Office
Title of the invention	**The Omni-Compass**
Inventor	Alan John Dubra, Kissimmee, Florida, United States
Drawing sheets	15
Application data sheet signed	20 September 2026 (form PTO/AIA/14, 37 CFR 1.76)
Publication	normal eighteen-month publication requested
Application number	withheld here on purpose; the filing is verifiable with the USPTO and is shown to a licensee under a signed Omni-Compass Enterprise License

The application data sheet names no assignee; the application stands in the inventor's name. Any assignment to The Omni-Compass LLC is recorded with the USPTO separately (37 CFR part 3).

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J.4 Trademarks

All patents, copyrights and trademarks covering the Omni-Compass engine, its mathematics and its software have been filed in the United States by The Omni-Compass LLC.

J.4.1 The marks

"Omni-Compass", "OmniCompass", the Omni-Compass compass rose with its ring of Greek letters, and the related names, logos and trade dress (the "Marks") are trademarks of The Omni-Compass LLC; trademark applications have been filed in the United States. Neither the evaluation license (`LICENSE`) nor publication of this repository grants any license to the Marks. A license to use them is granted only in a signed Omni-Compass Enterprise License, and only as it states.

J.4.2 What is allowed without a license

- Referring to Omni-Compass by name, accurately and in plain text, to say what it is, that you evaluated it, or that a result was produced with it, with the engine version and commit (`LICENSE`, section 4).
- Linking to the repository or to www.omni-compass.com.

J.4.3 What is not allowed

- Using any Mark, or anything confusingly similar, in the name of a product, service, company, domain, account, package, repository, fork or derivative.
- Using the logo or the compass rose, or altering any Mark.
- Suggesting endorsement, sponsorship, certification, partnership or affiliation by The Omni-Compass LLC.
- Using a Mark in advertising, comparative claims or sales material without a signed license.

J.4.4 Other companies' marks

Kubernetes is a trademark of The Linux Foundation; NVIDIA is a trademark of NVIDIA Corporation; KEDA and Karpenter are projects of their maintainers; other names belong to their owners. Their mention describes compatibility or a system Omni-Compass was tested with; it states no affiliation, endorsement or certification (`DISCLOSURES.md`, section 1).

J.4.5 Reporting misuse

The Omni-Compass LLC, main office infrastructure, toll-free direct 1-888-339-9551, system route asset 1-888-DEX-YLL-1, landline 321-746-4411 registered in Florida, www.omni-compass.com.

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APPENDIX

Appendix K. Security Policy

K.4 Reporting a vulnerability

Report a suspected vulnerability privately to The Omni-Compass LLC by telephone at its main office infrastructure (toll-free direct 1-888-339-9551; system route asset 1-888-DEX-YLL-1, landline 321-746-4411 registered in Florida) or through www.omni-compass.com, never in a public issue, pull request or discussion. Include the version and commit, what you saw, how to reproduce it, and its likely impact. We acknowledge reports, keep the reporter informed, and coordinate disclosure: a fix or a mitigation is published before the details are, and the reporter is credited if they wish. Please give us a reasonable time to fix before any disclosure, do not access data that is not yours, and do not disrupt any system; good-faith research that follows this policy will not be pursued by us.

K.5 What is in scope

The engine, the controllers, the plugs and wiring, the harnesses, the workflows and the verifier in this repository. Third-party software Omni-Compass runs on or connects to (Kubernetes, KEDA, Karpenter, NVIDIA drivers, databases) is reported to its own maintainers.

K.6 How Omni-Compass is built to fail safe

- The governor issues control actions only through the shield (`omnicompass/shield.py`, `cpp/src/shield.cpp`), inside the cover the operator granted, and never fights another writer.
- It snapshots every setting once, before its first write, and the kill switch (`python3 tools/omni_switch.py off`) hands every setting back; a dead-man lease hands them back if the governor stops.
- It runs under least privilege: on Kubernetes, a service account with `kubectl auth can-i` receipts for what it can and cannot do (`deploy/kind/rbac-omni.yaml`).
- The watchdog acts only on registry records its own user wrote that no other user can change, and never signals a process number the system gave to another process (`tools/omni_switch.py`). Put the switch and the registry in a directory only the governors' user can write (`OMNI_MASTER_OFF`, `OMNI_REGISTRY`); `status` warns when they are not.

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- It contains no telemetry and makes no network connection of its own except to the systems an operator points it at.
- The threats of misuse, the guards against each and what is still open are set out in `docs/THREAT_MODEL.md`.
- Every release carries a software bill of materials (`sbom/`) and a seal of the engine and its twins (`results/SEAL.json`, `python3 verify.py`).
- The capture script (`fleet/capture/kube_capture.sh`) is read-only; no component writes to a live cluster unless the operator starts it in a writing mode.

BACK MATTER

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Every term of the glossary where this book defines or discusses it, every muscle family where it is named, every file of the repository and every command-line setting this book names, with the pages they stand on. A plain mention of a common word is not an entry: the index points to where a thing is explained. A term whose pages would run past its line keeps the pages that fit, where it is defined and first used; after those it is throughout the book.

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BACK MATTER

Contact

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www.omni-compass.com

Repository: github.com/The-Omni-Compass-LLC/The-Omni-Compass

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